Archives of Computational Methods in Engineering (2025) 32:4197–4224
https://doi.org/10.1007/s11831-025-10264-1

REVIEW

Artificial Intelligence-Aided Design (AIAD) for Structures and Engineering: A State-of-the-Art Review and Future Perspectives

Yu Ao[1] · Shaofan Li[2] · Huiling Duan[1]

Received: 19 October 2024 / Accepted: 28 February 2025 / Published online: 18 March 2025

Abstract
Even with the state-of-the-art technology of computer-aided design and topology optimization, the present structural design still faces the challenges of high dimensionality, multi-objectivity, and multi-constraints, making it knowledge/experience-demanding, labor-intensive, and difficult to achieve or simply lack of global optimality. Structural designers are still searching for new ways to cost-effectively to achieve a possible global optimality in a given structure design, in particular, we are looking for decreasing design knowledge/experience-requirements and reducing design labor and time. In recent years, Artificial Intelligence (AI) technology, characterized by the large language model (LLM) of Machine Learning (ML), for instance Deep Learning (DL), has developed rapidly, fostering the integration of AI technology in structural engineering design and giving rise to the concept and notion of Artificial Intelligence-Aided Design (AIAD). The emergence of AIAD has greatly alleviated the challenges faced by structural design, showing great promise in extrapolative and innovative design concept generation, enhancing efficiency while simplifying the workflow, reducing the design cycle time and cost, and achieving a truly global optimal design. In this article, we present a state-of-the-art overview of applying AIAD to enhance structural design, summarizing the current applications of AIAD in related fields: marine and naval architecture structures, aerospace structures, automotive structures, civil infrastructure structures, topological optimization structure designs, and composite micro-structure design. In addition to discussing of the AIAD application to structural design, the article discusses its current challenges, current development focus, and future perspectives.

1 Introduction

1.1 Background

Structural design [1, 2] serves as the bedrock of engineering in different branches of engineering, as shown in Fig. 1, harmonizing stability, safety, efficiency, durability, functionality, aesthetics, and innovation for complex engineering systems such as shipbuilding, aerospace, automotive, and civil structures, and advanced engineering systems such as topological and composite structures, involving an in-depth analysis and refinement of structural load, material, element, model to ensure optimal design. Recently, as technology advances, higher performance expectations, interdisciplinary integration, and stringent standards, emerging engineering products have grown in complexity [3], transforming the related structural design process into a remarkably intricate and demanding process [4]. The current structural design tasks are primarily characterized by three characteristics:

- High-dimensional: Current structural designs involve a vast array of design variables, requiring complex and multidimensional optimization models.
- Multi-objectives: Current structural design involves comprehensive consideration of multiple objectives, requiring multidisciplinary collaboration.
- Multi-constraints: Current structural design involves numerous constraints, requiring integration of advanced optimization algorithms to overcome the highly discontinuous design space.

The characteristics of high-dimensional, multi-objectives, and multi-constraints faced by current structural design tasks result in the design process being knowledge-intensive and labor-intensive [5]. The heightened complexity and labor

✉ Shaofan Li
shaofan@berkeley.edu

1 College of Engineering, Peking University, Beijing, China

2 Department of Civil and Environmental Engineering, University of California, Berkeley, Berkeley, CA 94720, USA

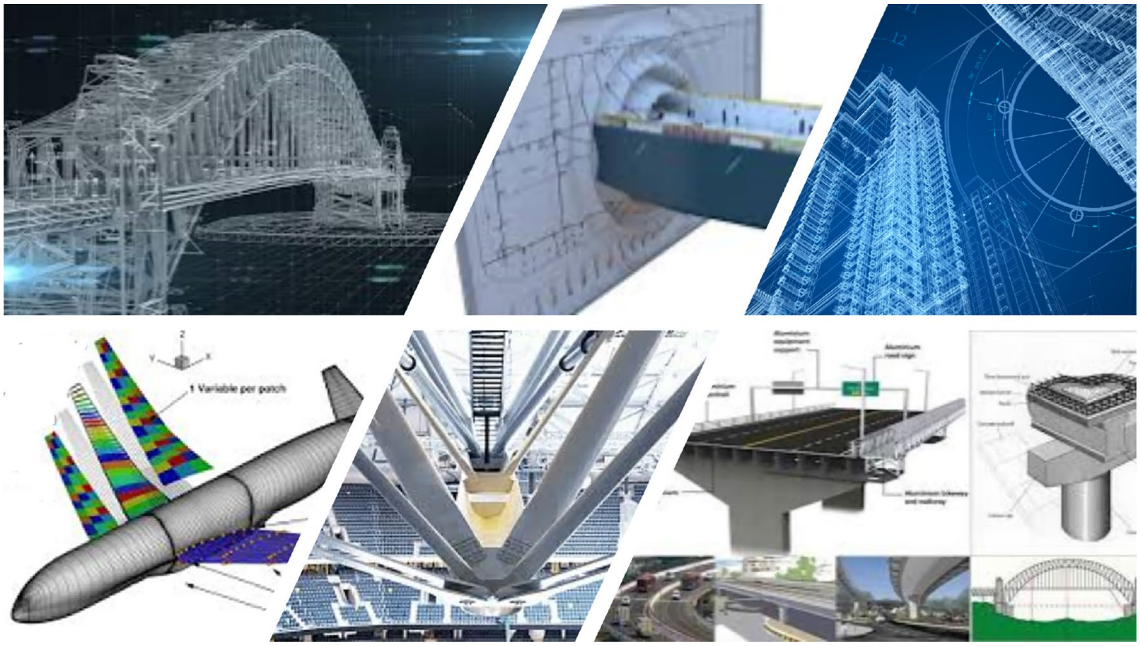

Fig. 1 Structural design and structure engineering

intensity experienced by designers have compelled them to actively pursue a novel, efficient methodology that can effectively support and expedite the structural design process.

1.2 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) [6, 7], a transformative field in computer science, has gradually been integrated into various fields since its conceptualization, bringing revolution and driving innovation. The early inklings of AI can be found in Greek mythology and philosophy, in which the idea of intelligent automatons and artificial beings was explored. AI as we are currently familiar was formed with the emergence of computers and the development of computing and intelligence theories, which began in the 20th century. In 1950, Alan Turing [8] proposed the idea of using machines to simulate human intelligence and introduced the concept of the Turing Test as a standard for evaluating machine intelligence. Subsequently, at the Dartmouth Conference in 1956, the term 'Artificial Intelligence' was coined, marking a foundational event in the field of AI research [9].

Although AI technology has developed over an extended period, early AI algorithms, such as Expert Systems [10] and Logic Programming [11], relied on manual programs that were time-consuming to create and difficult to adapt to new or unforeseen situations. The early feature of difficult to construct and low adaptability limited their scalability and practical utility, hindering the achievement of widespread applications. The emergence of Machine Learning (ML) [12] technology has significantly advanced the flexibility and applicability of AI. As a subject of AI, ML can learn from data, and make informed decisions based on the learned insights, rather than relying on explicit programming [13]. The learning and adapting ability without explicit programming shifts the burden from manual coding to algorithmic learning, enabling AI to handle complex tasks accurately and efficiently [14].

The relationship between AI and ML, along with commonly used ML algorithms, is illustrated in Fig. 2. As shown in Fig. 2, ML is usually divided into four categories: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. Supervised learning [15, 16] utilizes labeled data to develop mapping relationships between inputs and outputs, making it well-suited for performance prediction and regression analysis tasks. Unsupervised learning [17] learns from unlabeled data and identifies underlying patterns, which is ideal for data classification and anomaly detection tasks. Semi-supervised learning [18] examines labeled and unlabeled data simultaneously, using the labeled part to guide the learning process and the unlabeled part to improve performance, which is applicable when data labeling is costly. Reinforcement learning [19] implements reward and punishment mechanisms to enable the acquisition of physical patterns, beneficial in dynamic environments such as autonomous driving. Currently, a substantial proportion of AI algorithms in use are ML, with applications spanning various domains such as engineering, finance, and medicine.

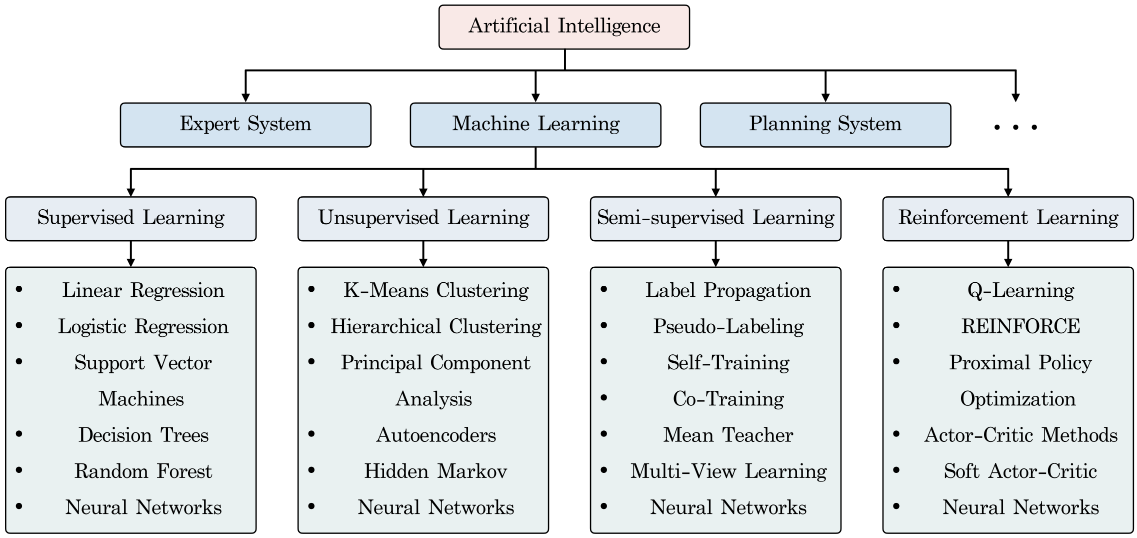

Fig. 2 Artificial intelligence and machine learning

1.3 Artificial Intelligence-Aided Design

The evolution of AI serves as a catalyst for innovation within the field of structural design, where researchers continue to explore its potential to optimize processes and improve outcomes. The initial surge of AI-driven structural design research can be traced back to the late 1980s, a period characterized by the emergence of Back-Propagation Neural Networks (BPNNs) [20]. Representative, Anluchene and Sun (1990) [21] employed BPNNs for concrete beam design and rectangular plate analysis. Hajela and Berke (1991) [22] investigated the feasibility of employing neural computing strategies in structural analysis and design. Ghaboussi et al. (1991) [23] explored the potential of utilizing neural networks for materials modeling. Thereafter, Reich et al. (1993) [24], Kang et al. (1994) [25], Adeli et al. (1995) [26], Mukherjee et al. (1995) [27], and Goel et al. (1996) [28] applied neural networks in various aspects of engineering design. Reich et al. (1997) [29] reviewed the application of ML to civil engineering problems and offered insights into its contributions to architectural design. Among the pioneers in AI-inspired design, MacCallum et al. [30–32] have conducted extensive research in the field of conceptual design, integrating knowledge-based empirical models. These earlier works provide valuable insights into the research of neural networks in the structural design of diverse domains, laying the groundwork for subsequent advancements in the field.

Early research in AI-driven structural design has shown considerable progress; however, constraints stemming from limited algorithms, computing power, and data accessibility led to early AI-based structural design research being primarily theoretical and with a restricted scope. Such challenges have been significantly mitigated with the development of Deep Learning (DL) [33]. Figure 3 presented the number of publications and citations from the Web of Science website when “AI”, “Structure” and “Optimization” or “Design” occur together. As shown in Fig. 3, since 2016, driven by breakthroughs in algorithms, increases in computational power, and the accumulation of data, DL, which evolved from ML, has experienced rapid development and has been gradually applied in different fields. Compared to traditional ML algorithms, DL algorithms possess much more complex layered structures and interconnected nodes [34], allowing algorithms to capture intricate and nonlinear patterns from high-dimensional data spaces, improving the predictive ability of AI algorithms when facing complex situations. The remarkable progress in algorithms promoted the progression of AI-driven structural design methods, enhancing their practical applicability and capacity to address complex tasks, culminating in the maturation of the Artificial Intelligence-Aided Design (AIAD) concept. As shown in Fig. 4, the AIAD-based structure design process comprises three key steps:

1. Data Preparation: Collecting data related to the structural design. The collected data should be able to reflect the relationship between structural design variables and structural performance. Common data collection methods include numerical simulations, physical experiments, and existing data collection.
2. AI Model Construction: Employing AI algorithms to perform pattern recognition on the mapping relationship between structural design variables and performance. Generating surrogate models to achieve low-cost prediction of structural design schemes. Commonly used AI algorithms include Support Vector Machine (SVM),

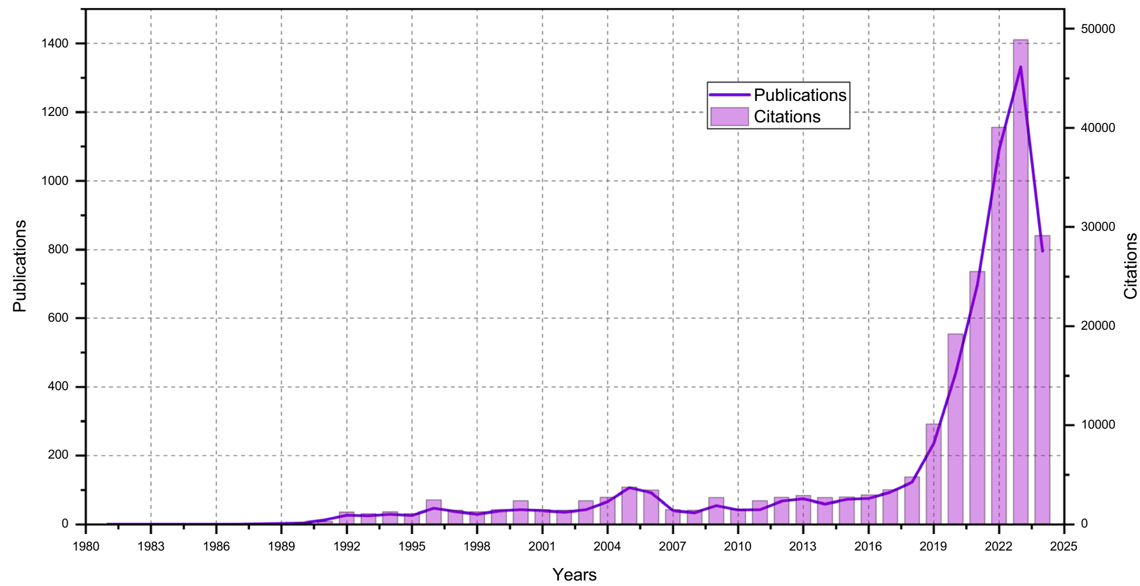

Fig. 3 Publication and citations of AI-based structure design by years

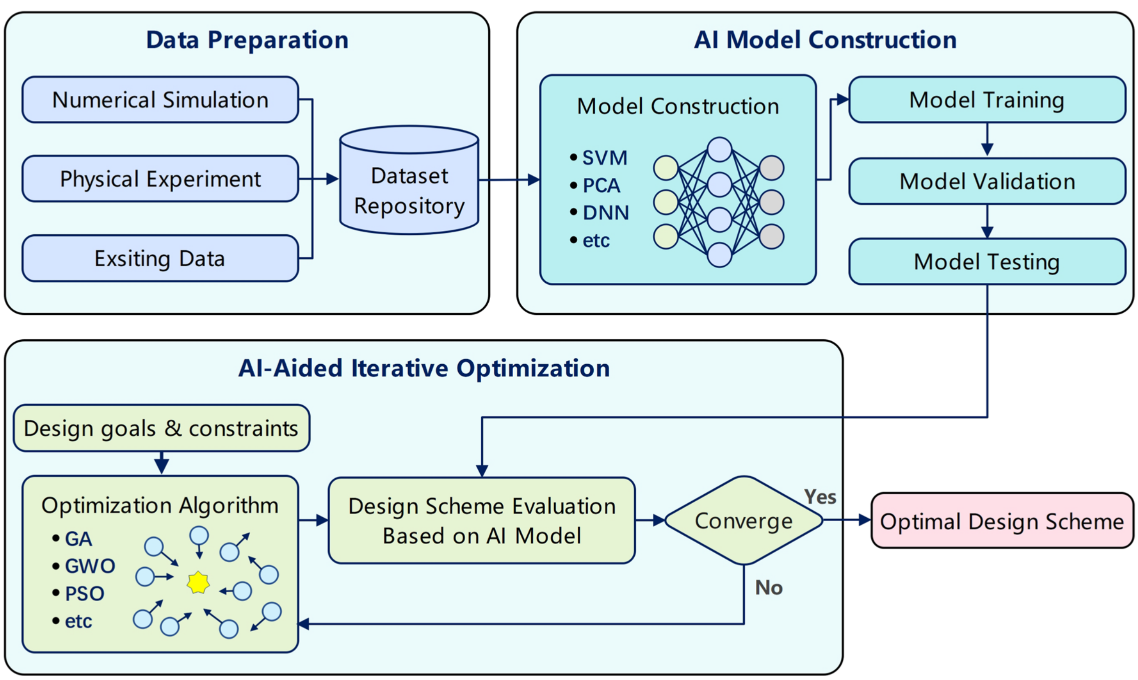

Fig. 4 Flow chart of the AI-aided design (AIAD)

Principal Component Analysis (PCA), Deep Neural Network (DNN), etc.

3. AI-Aided Iterative Optimization: Combining optimization algorithms with surrogate models to perform iterative optimization. Optimization algorithms are used for design space search, and surrogate models are utilized for fitness evaluation. Commonly used optimization algorithms include Genetic Algorithm (GA), Grey Wolf Optimizer (GWO), Particle Swarm Optimization (PSO), etc.

AIAD in structural engineering marks a paradigm shift in traditional design methods. Whereas conventional approaches typically advance from design conception to evaluation, AIAD employs a reverse methodology, initiating the design process by evaluating performance, then leveraging AI algorithms to judge the impact of structure on performance, and finally discerning optimal design parameters.

The paradigm shift introduced by AIAD effectively circumvented knowledge-intensive and labor-intensive challenges faced by traditional trial-and-error structural design. AIAD enables the design knowledge decoupling. In the traditional structural design process, knowledge is intensive and intertwined, requiring designers to not only have knowledge in the design field but also have a deep understanding of all disciplines (such as civil engineering, mechanics, materials, safety, etc.) involved in the structure index. Through the adoption of AIAD, this intensive and intertwined can be significantly alleviated, as AI algorithms can acquire physics knowledge across various domains through data-driven learning processes thereby equipping designers with insights and tools for informed scheme-evaluating. Meanwhile, AIAD enables the design labor reduction. In traditional structural design processes, each iteration requires significant time and effort to evaluate the new scheme, leading to increased labor costs and prolonged design timelines, particularly in high-dimensional and multi-disciplinary projects. AIAD employs AI to understand the mapping relationship between design variables and performance, then construct an AI surrogate model to assist designers in assessing the impact of design variables on structural performance, effectively simplifying the evaluation process and avoiding design labor explosion in high-iteration situations. Hence, leveraging AIAD not only advances structural design towards greater efficiency and simplification but also streamlines the design workflow.

1.4 Purpose of This Review

Given the exponential growth in AIAD applications for structural design in recent years, there have arisen an urgent need for a refreshed and updated review of AIAD's impact on structural engineering. To foster the development of AIAD for structural applications, this study undertook a systematic literature review focusing on AI-based engineering structure design research. The purpose of this work is to investigate the application potential of AIAD. To achieve this goal, we assessed the extent to which AIAD has been explicitly adopted for structural design applications in different disciplines. We also analyzed the similarities in the application of AIAD across various disciplines in structural design to determine the future direction of AIAD. Our research questions are formulated as follows:

- Q1: To what extent has AIAD been applied in structural design?
- Q2: What is the future roadmap for the adoption of AIAD in structural design?

To address these questions, this study employed the content analysis-based approach to conduct a comprehensive literature review during the review process. As a methodology comprising data analysis and interpretation, content analysis effectively guarantees the rigor and systematic nature of the literature review [35]. Six representative domains have been examined: shipbuilding structures; aerospace structures; automotive structures; civil structures; topological structures; and composite structures. In all these domains, shipbuilding, aerospace, automotive, and civil structures are classified as macrostructures, while topological and composite structures are classified as microstructures. For reading convenience and easy reference, we listed the nomenclature used in this article in Table 1.

The rest of this article is organized as follows. Section 2 discussed the review methodology. Section 3 provided an overview of AIAD research on the ship hull structures. Section 4 reviewed the application of AIAD in aerospace structures. Section 5 focused on the AI-based vehicle structure design. Section 6 offered an examination of AIAD research relevant to civil structures, encompassing building and bridge structures, as well as infrastructures. Section 7 explored AIAD research concerning the design of topological structures. Section 8 reviewed the research of AIAD for composite structure design. Section 9 concluded with a discussion summarizing the findings, limitations, and future development. Section 10 summarized the contents of the article.

2 Literature Review Methodology

The flow chart of the whole literature review process can be seen in Fig. 5. As shown in Fig. 5, the first step in conducting a literature review is the material collection, which is completed by keyword retrieval. Given that the purpose of this study is to outline the current extent and future roadmap for the application of AIAD in structure design, the keyword selection criteria for the material collection were as follows:

1. The articles should focus on the AI application in structure design. They should involve structure design or optimization and include at least one AI-related technology such as artificial intelligence, machine learning, deep learning, neural network, or data-driven technology.
2. The articles should belong to one of the selected representative engineering fields (shipbuilding, aerospace, automotive, civil, topological, and composite) and contain a related keyword in the title, keywords, abstract, or content.
3. The articles should be journal articles, and be indexed in the Science Citation Index (SCI), and Science Citation

Table 1 Nomenclature of methods

Abbreviation	Explanation	Abbreviation	Explanation
AI	Artificial Intelligence	AIAD	AI-Aided Design
GAI	Generative Artificial Intelligence	ES	Expert System
PS	Planning System	DM	Data Mining
SVM	Support Vector Machine	RF	Random Forest
DT	Decision Theory	DST	Dempster-Shafer Theory
ML	Machine Learning	ELM	Extreme Learning Machine
DL	Deep Learning	NN	Neural Network
ANN	Artificial Neural Network	DNN	Deep Neural Network
FCNN	Fully Connected Neural Network	BPNN	Back Propagation Neural Network
GAN	Generative Adversarial Network	PCA	Principal Component Analysis
GNN	Graph Neural Network	KADS	Knowledge-Aware Design System
KAE	Knowledge Aided Engineering	MD	Multidisciplinary Design
MID	Multidisciplinary Inverse Design	HOA	Heuristic Optimization Algorithm
GA	Genetic Algorithm	GWO	Grey Wolf Optimizer
PSO	Particle Swarm Optimization	BAS	Beetle Antennae search
CFD	Computational Fluid Dynamics	FEM	Finite Element Method
FEA	Finite Element Analysis	BEM	Boundary Element Method

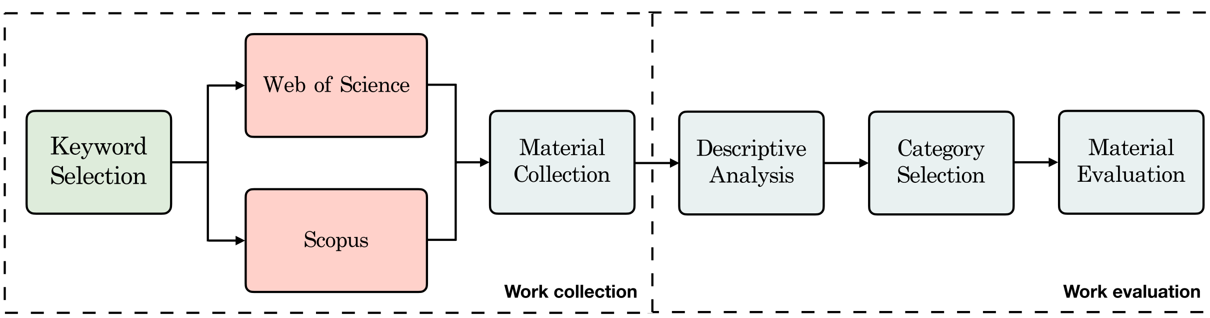

Fig. 5 Flow chat in the literature search process

Index Expanded (SCI-E) to ensure significant impact on target areas.

According to the selection criteria, the keywords for article collection are shown in Fig. 6. As shown in Fig. 6, the filter keywords consisted of three main categories: research content, research method, and research domain. The research content keywords ensured the filter articles focused on structure design or structure optimization. The research method keywords required the filter articles to use AI-related methods. The research domain keywords limited the filter articles to the six selected representative engineering domains covered in this review.

During the article collection process, keywords in each category were iteratively expanded in the order of content, method, and domain. In each iteration, a new keyword was incorporated into the keyword combination, accounting for synonyms, thereby maintaining the completeness of search results. Two academic search engines were used to conduct the material collection, including Clarivate Analytics's Web of Science, and Elsevier's Scopus. After searching through three engines to obtain the dataset of articles that met all three requirements, duplicate search results were removed.

Based on the collected materials, we then conducted the descriptive analysis by analyzing the bibliographic information of the collected articles. The information for the collected articles gathered encompassed a variety of bibliographic metrics, including the title, abstract, author(s), journal, year of publication, number of citations, and article type. Based on the bibliographic information, the collected articles in different domains were categorized based on content and research methodology. In this article, we mainly filtered case, simulation, and modeling studies, and focused on their application scenario. By completing the descriptive analysis and category selection, we obtained insights into the evolution of research focus within the domain and reveal shifts in research priorities and the emergence of new areas of interest.

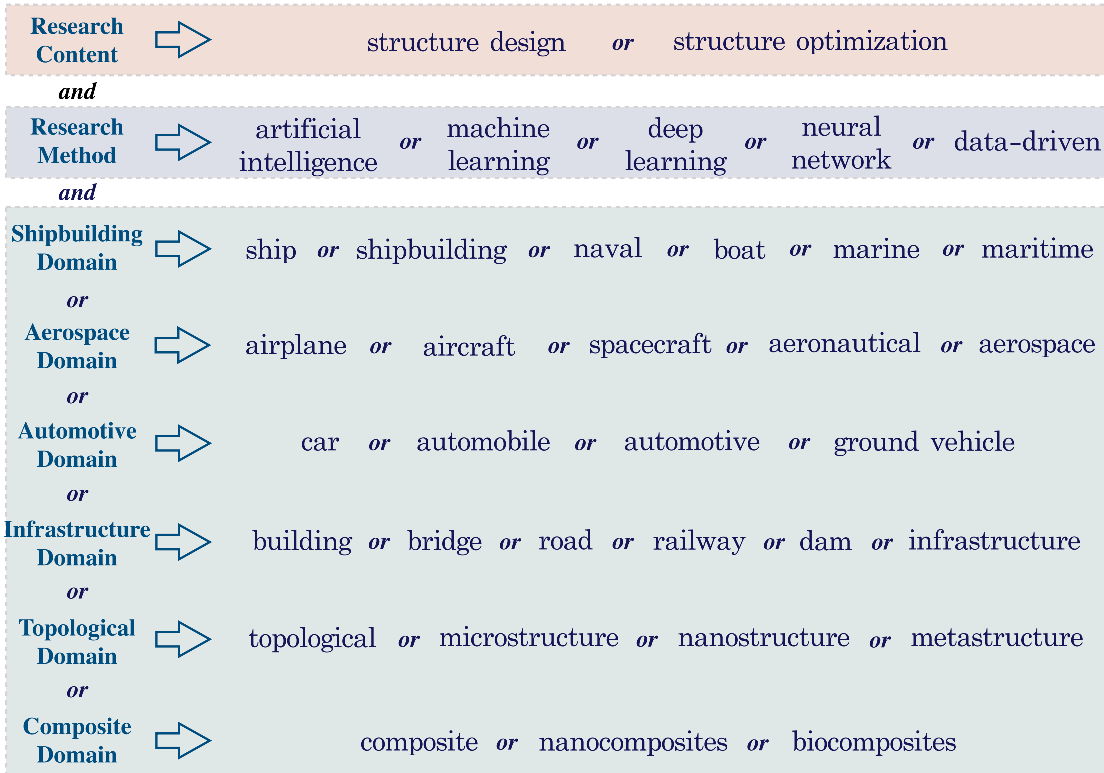

Fig. 6 Keywords for the article collection

To improve the relevance of the collected articles to the review topic, we then conducted the material evaluation process after the category selection. In the material evaluation process, we judged the articles' relevance by reading their abstracts. When the abstract of an article explicitly mentioned the use of AI algorithms to complete the optimization, design, or performance prediction of certain structures, the article was judged as a highly relevant article.

3 AI-Aided Designs for Ship Hull Structures

The ship hull structure design constitutes a pivotal phase in shipbuilding, bearing importance to the construction and performance of vessels. Current ship hull structure design tasks face stringent challenges due to enhanced market competition and safety regulations. These contemporary challenges are marked by complex and interrelated variables, objectives, and constraints, necessitating innovative solutions and adaptive strategies to meet these multifaceted demands effectively. Due to its ability to address the challenges of multi-variable, multi-objective, and multi-constraint, AIAD is increasingly being adopted in the ship hull structure design field. After aggregating and removing duplicates of the papers retrieved from each database using the search keywords presented in Fig. 6, the total number of collected articles amounted to 1414.

The initial endeavor to incorporate AI-related methodologies in ship hull structure design dates back to 1996. Ray et al. (1996) [36] demonstrated the effectiveness of neural networks in naval architecture and marine engineering, with two application examples of container capacity prediction and added mass coefficient estimation. Their work highlighted the potential of AI to handle complex, nonlinear problems, enhancing design and analysis in marine applications. Since then, scholars have continued exploring AI technology's application in ship structure design. Yang and Sen (1997) [37], rooted in Decision Theory (DT) and the Dempster-Shafer Theory (DST), offered a structured framework for selecting design options and demonstrated the effectiveness through a real-world design selection problem. In addition, Reich (1998) [38] illustrated the ML method in the marine propeller design decision-support system. Alkan and Gülez (2004) [39] develops a NN-based prediction tool for ship stability and load capacity, enhancing design efficiency and safety. Ramazani et al. (2013) [40] utilized ANN to estimate wave-induced pressure loads on marine composite panels.

However, due to data volume, computer power, and algorithm capacity limitations, early AI algorithms exhibited low accuracy in predicting performance, resulting in limited applicability of AI technology in ship structures, leading to few related articles being published annually over an extended period. Such limitations have been a breakthrough with the development of AI technology, especially after 2016, significant advancements in AI technology have markedly transformed AIAD for ship hull structure, substantially enhancing its applicability. The development of AI technology has made the application of AIAD in ship structures more sophisticated and practical. Representative, Li et al. [41] employed the NN to realize structure material collaborative optimization for underwater vehicle composite pressure hulls. Ahn et al. [42] created an experimental database for LNG tank sloshing loads and used an ANN for load prediction, emphasizing the possibilities of the database-based cargo hold design.

In 2021, the present authors [43] introduced the concept of AIAD in the field of ship hull structure design. As presented in Fig. 7, the authors employed the Fully Connected Neural Network (FCNN) model to predict the total resistance of ship hulls, governed by a series of control points, Building upon the initial research, the author undertook an extensive series of studies [44–46], to investigate the effectiveness of AIAD in improving design precision, streamlining construction workflows, and enhancing operational efficiency under diverse application scenes. Figure 8 illustrates the design space of the bulbous bow for container ship hull

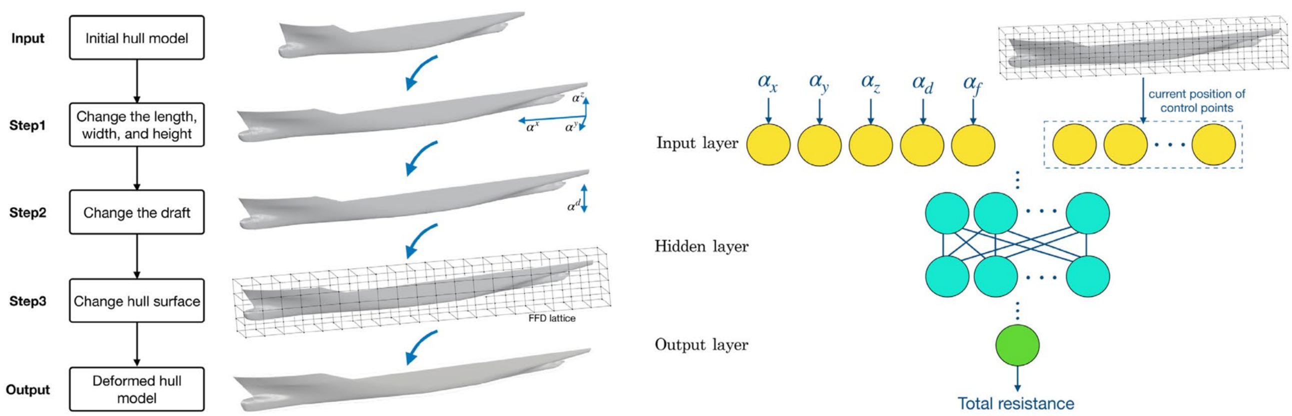

Fig. 7 Hull parameterization and FCNN-based resistance prediction of Ao et al. (2023) [43]

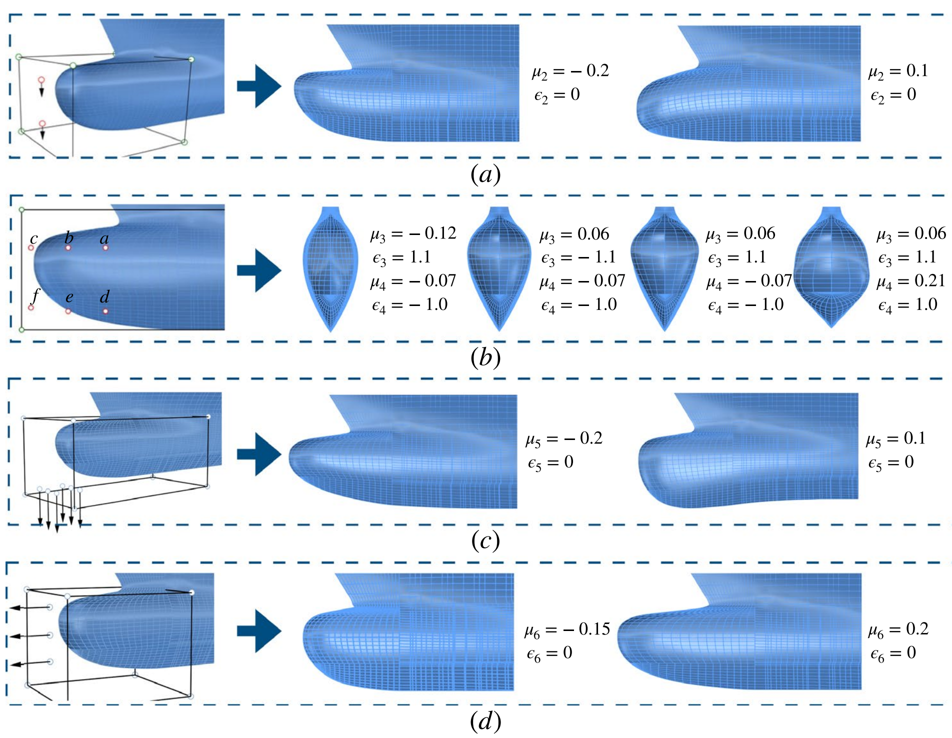

Fig. 8 Control points and design space for hull bulbous bow of Ao et al. (2024) [46]. **a** Deformation of the bulbous bow longitudinal section. **b** Deformation of the bulbous bow cross-sectional shape. **c** Deformation of z-direction longitudinal shape of the bulbous bow. **d** Deformation of x-direction longitudinal shape of the bulbous bow

structures, demonstrating the robust capabilities of AIAD in managing and optimizing complex design objects. Similar AIAD research has also been conducted with varying design objectives and conditions on ship hull structures.

Huang et al. [47] provided an overview of ML in enhancing sustainability in the shipping industry, and discussed ML applications in ship design. Pablo [48] introduced an ANN based tool for early-stage seakeeping prediction in ship design, offering speed and accuracy without needing precise hull shapes. Building on their prior research, Pablo et al. [49, 50] conducted a series of experiments utilizing ANN to predict the seakeeping qualities of ships at forward speed, further highlighting the potential of AI in naval architecture. Bagazinski et al. (2023) [51] utilized Generative Artificial Intelligence (GAI) to achieve high-performance parametric ship hull design. The overview of the ship hull generation process of Bagazinski et al. is given in Fig. 9. Following this, they aggregated a comprehensive dataset named Ship-D [52], comprising 30,000 ship hulls along with essential design and functional performance data, thereby offering a crucial resource for advancing data-driven ship design further. Tripathi et al. (2023) [53] implemented a DNN surrogate model for designing stern flaps on high-speed ships, significantly reducing the time and cost associated with traditional testing methods.

In conclusion, traditional ship structures require evaluating performances including cabin structures and hull structures, resulting in labor-intensive and time-consuming processes. AIAD achieves fast optimization and design by integrating AI algorithms, even when facing complex fluid-solid coupling problems. Hence, AIAD has the potential to be applied in ship structure design tasks with multiple design variables, objectives, and constraints, which are typically time-consuming in the traditional design process, thus enhancing the generalization of the design result.

4 AI-Aided Design for Aerospace Structures

The structural design of aerospace vehicles, including airplanes and space vehicles, is critical to developing these advanced systems, significantly impacting their performance, safety, and reliability. Aerospace structure design is characterized by unique challenges such as the need for lightweight yet robust materials, and the ability to withstand extreme aerodynamic heat, pressure, and vibrations. Similar to the shipbuilding structure design, traditional iterative design methods for aerospace and space structures often depend on physical testing or simulation with a number of iterative processes, limiting the potential to explore complex and non-traditional design solutions. The emergence of AIAD ideas gradually addresses the shortcomings of traditional design methods, greatly simplifying the process of the aerospace structure design. Based on the search keywords, the total number of collected articles amounted to 1506, and the time scale spans from 1991 to 2024.

Although not diretly mentioned, the first trace of utilizing AIAD concepts in aerospace structures can be found in 1992, when Umaretiya and Joshi (1992) [54] mentioned the potential application of a knowledge-based system for aerospace structure design. Since then, scholars have conducted a series of studies. Rokhsaz et al. [55] discussed the

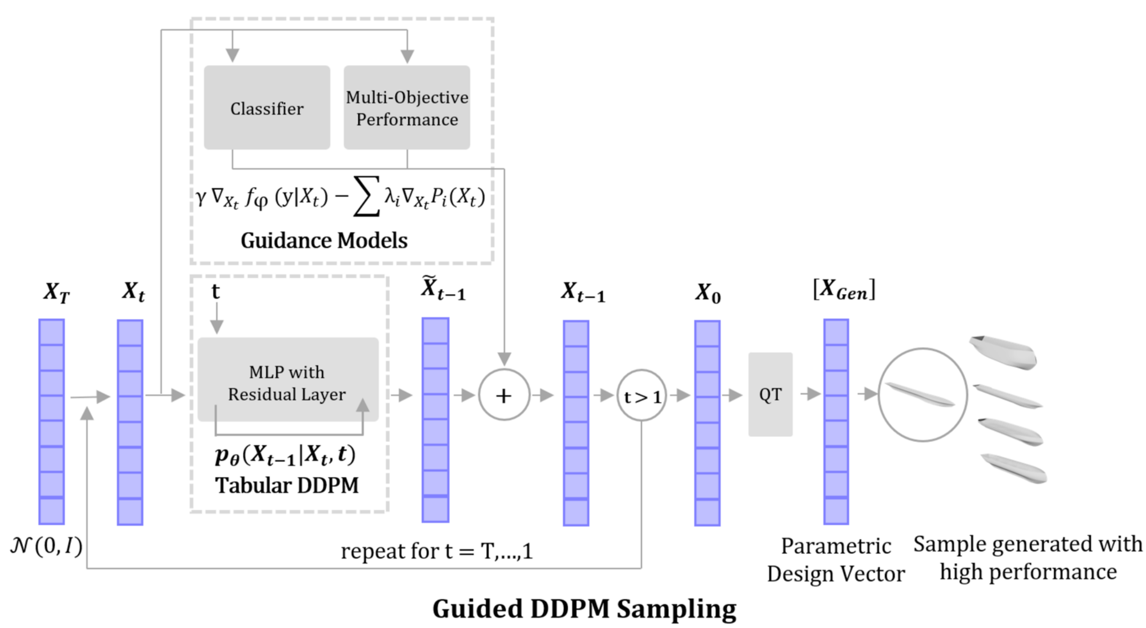

Fig. 9 Process of the ship hull parametric generation based on GAI of Bagazinski et al. (2023) [51]

applications of AI in nonlinear aerodynamics and aircraft design. Giannakoglou (2002) [56] employed ANNs as surrogate models to assist in the design of airfoil shapes, reporting a more efficient and effective stochastic optimization. The schematical presentation of the airfoil shape design process of Giannakoglou is given in Fig. 10. Then, Forrester et al. [57] further explored the benefits of surrogate models for aerodynamic-based engineering optimization. Park et al. (2013) [58] proposed a NN-enhanced reduced-order model for aerostructure optimization of wings, decreasing the number of experiment designs. Lindhorst et al. (2014) [59] adopted a surrogate model to achieve the nonlinear aeroelastic analysis of the NLR7301 airfoil, significantly reducing computation time. However, limited by the capability constraints of the algorithm used to build the surrogate model, early design objects and conditions were relatively idealized. Take the AI-based wing design for instance, most articles only considered two-dimensional contours, without considering the entire wing and key characteristics such as the wingtip vortex.

The advancements in AI algorithms have greatly enhanced the development of AIAD in aerospace structures, transforming the application scenarios to be more complex and diverse. Representative, Garriga et al. (2019) [60] focused on the electrification of flight control and landing gear systems, and developed an integrated design framework for aircraft systems using a combination of simulation tools, ML-based clustering, and optimization to efficiently analyze and optimize various system architectures. Pitton et al. (2019) [61] optimized the buckling resistance of aerospace thin-walled cylindrical shells by combining AI with the Finite Element Method (FEM), achieving a 4% increase in buckling load. Dong et al. (2021) [62] provided a comprehensive review of how DL techniques have been and can be applied to aircraft design, dynamics, and control, highlighting various methods and discussing future opportunities and challenges in the field.

Subsequently, Agarwal et al. (2022) [63] employed the Finite Element Analysis (FEA) method and AI techniques to analyze deformation in AA2024, demonstrating how different channel and corner angles affect stress, strain, and temperature in aircraft structural materials. Yong-fei et al. (2023) [64] employed ML to construct a Knowledge-Aware Design System (KADS) aimed at the rational design of

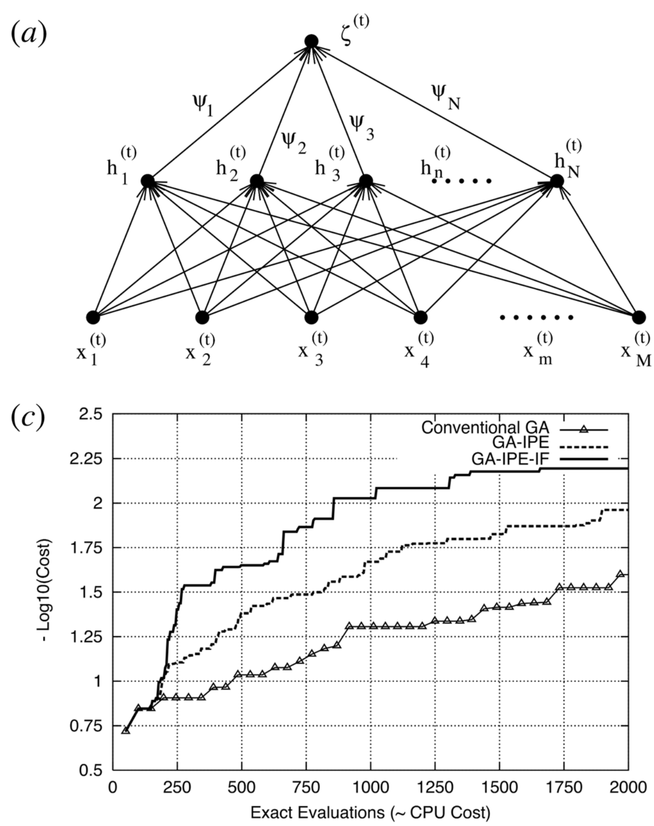

Fig. 10 Schematical presentation of the airfoil shape design method of Giannakoglou (2002) [56]. **a** The RBFN with a single-output unit. **b** Schematic diagram of the airfoil shape design process. **c** Comparison of optimization convergence speed with and without ANN. **d** Re-design of the NACA 0012 profile: best computed profile, optimal location of the Bezier points and search spaces. **e** Re-design of the NACA 4412 profile: best computed profile, optimal location of the Bezier points and search spaces

high-strength aviation aluminum alloys. Their AI-based design methodology facilitated the efficient design and fabrication of an aluminum alloy exhibiting mechanical strength. Dede (2024) [65] proposed an ML-aided generative design methodology for the Mars regolith habitation shell, integrating generative design, simulation tools, and data science methods, and realized the design of a habitation shell that can withstand the Mars harsh conditions. Al-Haddad and Mahdi (2024) [66] developed an FEA and ML-based simulation method for aircraft landing gear, allowing for adaptable, quick simulations while ensuring accuracy.

AIAD also contributes to aerospace and space shape structure design. Li et al. (2021) [67] introduced the data-based approach to the high-fidelity wing design and achieved optimization for common research wing models. Renganathan et al. (2021) [68] realized a DNN-based highly parameterized wing airfoils design. Liu et al. (2022) [69] optimized a morphing wing with the help of the Kriging surrogate model. Wu et al. [70] optimized the aerodynamic shape of a missile using an optimization framework composed of four intelligent algorithms. Taj et al. (2023) [71] presented a Multidisciplinary Design (MD) framework optimizing aerodynamics and radar cross section for fighter aircraft based on ML.

In addition, Zhang et al. (2024) [72] extended the application of AI to the design of three-dimensional airfoils, where the design space is determined as shown in Fig. 11. As presented in Fig. 12, they utilized the adaptive DNN-based optimization approach, enabling the production of optimal wing configurations for various aerial vehicles under diverse cruise conditions. Sharma and Hosder [73] demonstrated the feasibility of the ANN-based inverse design of blended wing

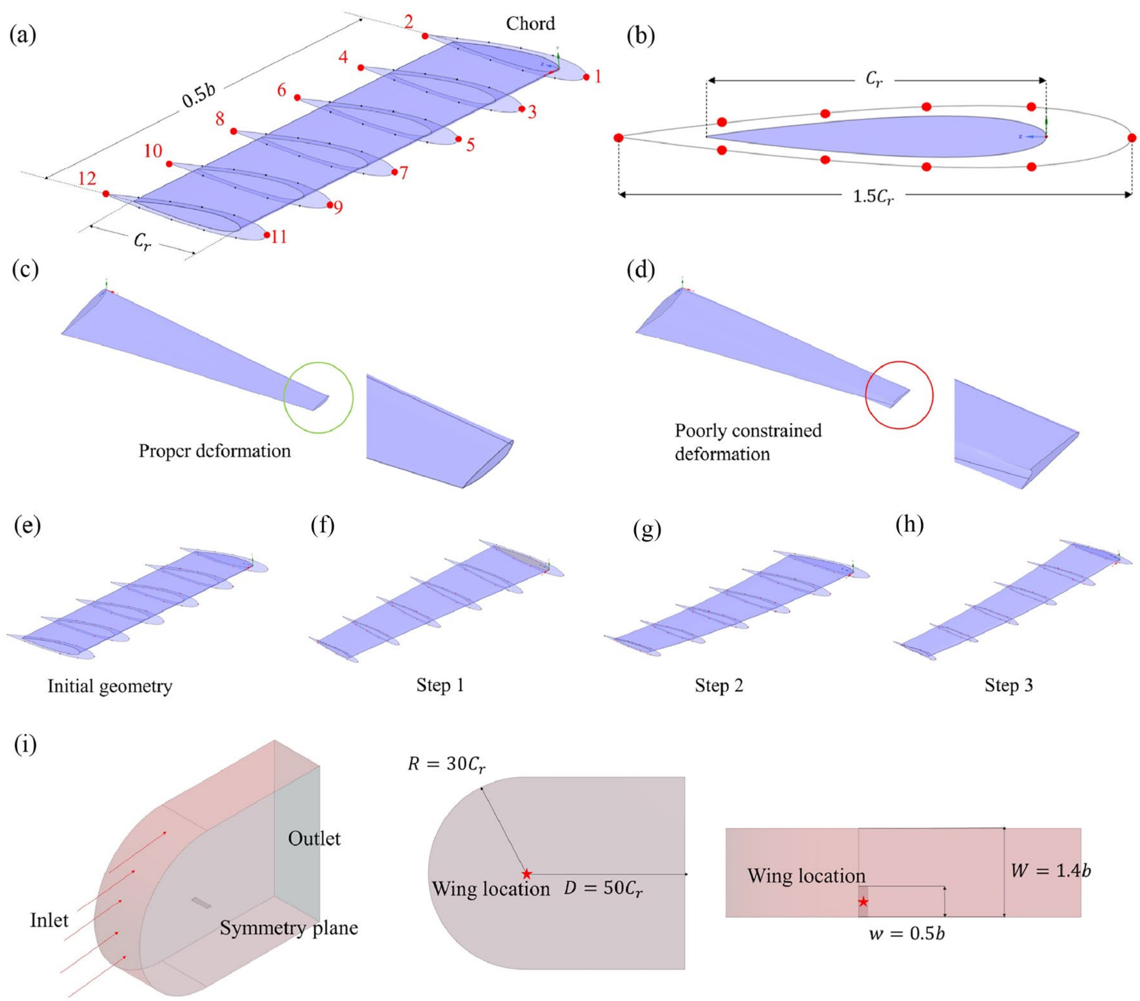

Fig. 11 Search for aeroplane wing design space: Control points distribution and deformation of Zhang et al. (2024) [72]. **a** Perspective view showing partially labeled control points at the leading and trailing edges for constraint purposes. **b** Cross-sectional view. **c** Example of acceptable deformation. **d** Example of poorly constrained deformation. **e** Initial NACA 0012 wing configuration. **f** Step 1: Linear deformation. **g** Step 2: Vertical and downstream cubic deformation. **h** Step 3: Deformation led by local control points. **i** Fuid domain dimensions for the CFD simulation

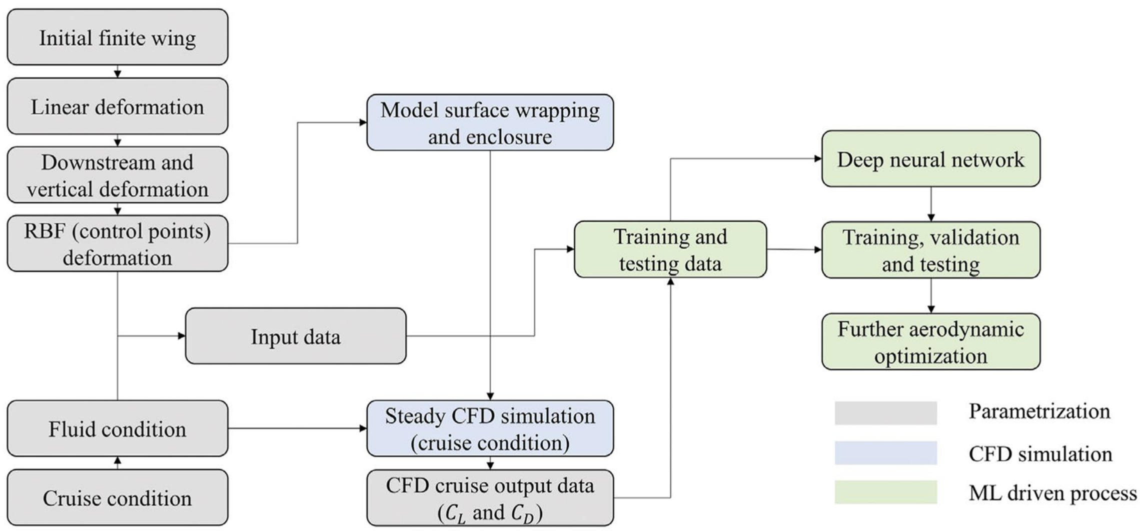

Fig. 12 Flowchart of wing profile design process combining DNN and CFD of Zhang et al. (2024) [72]

body aircraft by predicting configuration design variables from mission-informed performance data. Except for wing-shape design, researchers have also applied AI to the design of other aerodynamic-related structures. For instance, Karali et al. (2024) [74] introduced an AI-driven MD method for the initial dimensions of key components of Unmanned Aerial Vehicles (UAVs).

It can be seen that AIAD is also widely used in aerospace structures due to its high degree of customizability. By combining different numerical simulation methods, AIAD enables the design of aerospace structures aimed at improving aerodynamic performance or structural strength. Currently, in the aerospace structure domain, the majority of applications of AIAD primarily focus on the design of aircraft wings or critical component structures. In subsequent studies, by increasing data samples and enhancing AI algorithm complexity, AIAD could be extended to multiple aerospace structures.

5 AI-Aided Design for Automotive Structures

Similar to ships or airplanes, automotive structure design mainly focuses on enhancing safety, fuel efficiency, and manufacturing process optimization. In addition, being commercial products, cars are also prioritized to improve user experience as one of the main goals of structural design. Scholars have conducted a series of studies on AI-aided automotive structural design. Based on the search keywords, the total number of retrieved articles related to the automotive structures amounted to 464.

The first instance of directly integrating AI into automotive structures can be traced back to 2000 when Susca et al. (2000) [75] introduced a Knowledge Aided Engineering (KAE) system for racing car design. Then, Bae et al. (2003) [76] proposed a NN-based method to predict fatigue design criteria for spot-welded structures of automobile and train bodies. Compared to that in shipbuilding and aerospace, early research on the design of automobile structures using AI is less extensive. This is mainly due to the large-scale production model of the automotive industry, which requires more technological accumulation to drive innovation. Nowadays, with the development of AI algorithms, the automotive industry is increasingly recognizing AI's potential for vehicle structural design and has been progressively applying these technologies to improve the structural design process. In 2018, Journal Advances in Mechanical Engineering (AIME) organized a special collection *Machine learning in automotive industry* [77], presenting ten cutting-edge articles that showcase recent advancements and the diverse application potential of ML in the automotive industry, reflecting the application trend of AI and ML in the automotive industry.

A critical research field of AIAD in automotive structure design is the enhancement of safety. AI methodologies are utilized to improve the efficiency of vehicle crash scenario modeling, forecast potential points of failure, and refine the structural strength and integrity of vehicles to ensure occupant protection during impacts. Representative, Wang et al. (2002) [78] utilized the BPNN-based deformation design method to accomplish the automotive front bumper model design, improving efficiency by 55.71% over manual methods. Kim et al. (2022) [79] demonstrated the efficiency of a DL-based Multidisciplinary Inverse Design (MID) for automotive brake optimization. The brake design process of their work is given in Fig. 13. Based on their work, Kim et al. concluded that the DL-based MID method can balance apparent piston travel and drag torque, which are often treated as conflicting requirements.

In addition, Wang et al. [80] combined quantum PSO with an extreme learning machine (ELM) and realized the prediction of wheel tread wear in heavy haul freight cars and thereby optimized wheel profile with low wear. Al-Shabi et al. (2023) [81] employed the ML algorithm and the GWO algorithm to design the vehicle door to minimize weight under the side impact condition while maintaining structural safety. Töpel et al. (2023) [82] developed a design optimization method using ANNs to enhance the design of chassis bushings, improving their stiffness characteristics through FEM simulations and the PSO algorithm. Cheng et al. (2024) [83] presented a DNN-based model to simulate the transient dynamics of a car's hood door under axial mechanical shock loading during an accident, emphasizing its application in structural design.

Beyond safety considerations, fuel efficiency is also a pivotal research field of AIAD in automotive structural design. AI algorithms aid engineers in efficiently and comprehensively refining vehicle structural shapes to enhance aerodynamic efficiency, thereby lowering fuel consumption and reducing emissions. Representative, Cao et al. (2023) [84] introduced AI into automotive aerodynamics. With the help of the U-net network shown in Fig. 14, they proposed an AI-based fast CFD simulation model for automotive aerodynamics, significantly shortened the automotive aerodynamic profiling design cycle. Mohamed et al. (2024) [85] introduced DrivAerNet, a comprehensive CFD dataset for car shapes, and a graph-based NN model for predicting aerodynamic drag, enabling rapid and accurate aerodynamic assessment during the car design process.

In addition, improving user experience is another key research area of AIAD in automotive structural design. By leveraging AI-driven methods, engineers can efficiently and creatively optimize structure layouts and enhance comfort. For instance, Ma et al. (2022) [86] introduced a hierarchical model for the recognition of car front-facing styles, which can automatically categorize them into eight distinct types. Ding et al. (2023) [87] combined the BPNN and SVM model with FEM and Boundary Element Method (BEM) simulations to optimize the door inner panel structure, achieving the quality of the car door closing sound enhancement.

Compared with shipbuilding and aerospace, AIAD is less frequently utilized in automotive structural design, primarily due to the commercial attributes of automobiles. With advancements in AI technology, AIAD's potential has been noticed and has started applied to automotive structural design. Currently, AIAD is already being utilized to improve structural strength, reduce wind resistance, and enhance user experience. In the future, AIAD could evolve into a new method for automotive structural design.

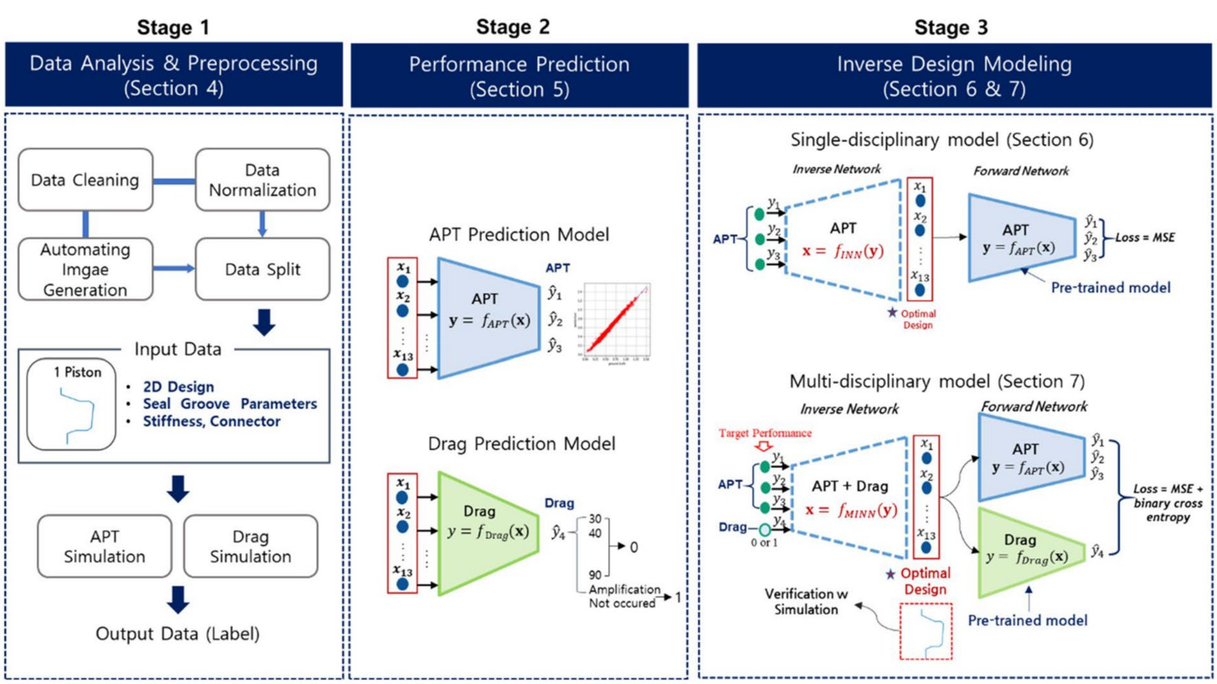

Fig. 13 DL-based brake systems inverse design process of Kim et al. (2022) [79]

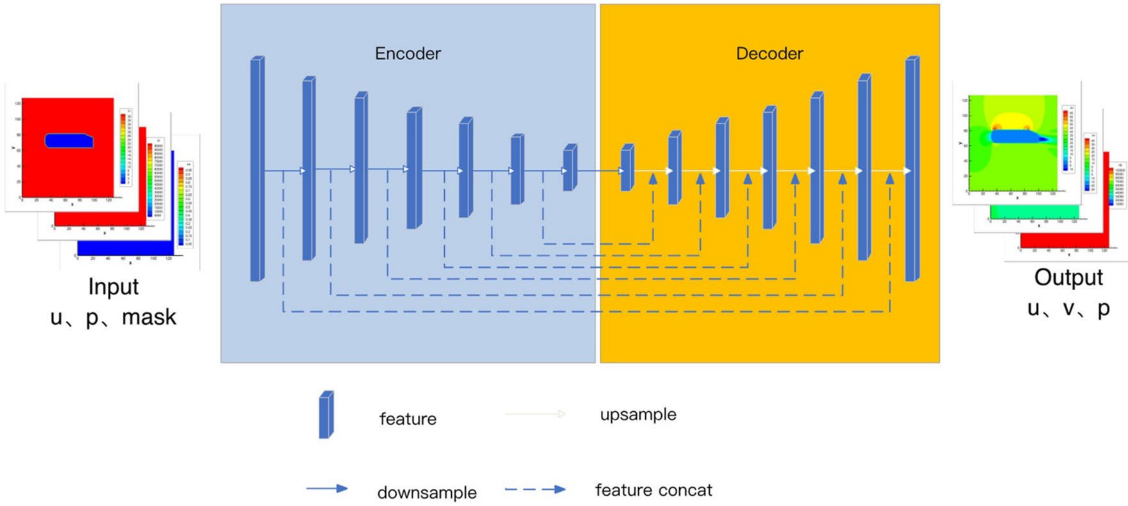

Fig. 14 Neural network architecture for automotive aerodynamics prediction of Cao et al. (2023) [84]

6 AI-Aided Design for Civil Structures

Civil structures, including buildings, bridges, and other infrastructure structures, are fundamental to modern society. Design objectives for these structures can be classified into ensuring structural safety, promoting material performance, and enhancing aesthetic appeal. Given their complexity and scale, a multidisciplinary approach is essential, integrating principles from structural evaluation and material engineering, resulting in the cumbersome of the overall design process. Combined with AI technology, scholars have conducted numerous studies in the field of civil structure design. Based on the search keywords, the total number of retrieved articles related to the civil structures amounted to 4887.

In the realm of civil structural design, the application of AI technology has a long-standing history and broad implementation. Early efforts to integrate AI technology into civil structural design attempted a diverse array of AI algorithms. Notably, as early as 1991, Miles and Moore [88] developed an Expert System (ES) for bridge conceptual design, capturing expert engineers' knowledge to aid inexperienced designers. Reich and Fenves (1995) [89] further explored the use of ML in bridge preliminary design, emphasizing ML's ability to generate, update, and use preliminary design knowledge. Seyedpoor et al. (2012) [90] proposed an efficient optimization method for the shape design of materially nonlinear arch dams based on the BPNN, accounting for dam-water-foundation rock interaction during earthquakes, where the design target and the flowchart are presented in Fig. 15.

Similar to other domains, benefiting from the rise of DL, the research field of AI-assisted civil structure design has progressively moved towards the application of multi-layer ML and DL. This evolution not only guarantees the optimization accuracy of AIAD but also enhances its problem-solving capabilities. Akinosho et al. (2020) [91] reviewed the application of DL in addressing challenges within the construction industry, highlighting its untapped potential in areas such as site planning, health and safety, and design. Xie et al. (2020) [92] reviewed the transformative role of ML in earthquake engineering, providing insights into the application of ML in seismic hazard analysis, system identification, damage detection. Málaga-Chuquitaype (2022) [93] reviewed the evolution and impact of AI and ML in building structural engineering, discussing current applications, challenges, and the potential diminishing need for human engineers in conventional design practice. Afshari et al. (2022) [94] reviewed the development and application of ML techniques to enhance the efficiency and accuracy of structural reliability analysis in civil and mechanical engineering. Ocak et al. [95] reviewed the application of AI, particularly DL, in civil engineering, and discussed how AI mimics human cognition to solve complex problems by enabling predictive models, highlighting AI's potential for enhancing engineering solutions.

One primary objective for civil structural design is to enhance performance, particularly in terms of safety and stability. In this context, AIAD is utilized to design the structural frame or components of buildings, bridges, and infrastructure projects, optimizing their strength performance and structure integrity. Representatively, Geyer and Singaravel (2018) [96] presented a component-based approach for ML in building cooling and heating performance prediction, enhancing the building design for both whole-building and component-level. Liu et al. (2020) [97] developed a ML-based approach to design and optimize curved beams with varying thicknesses, achieving improved performance in bistable structures, validated through experimental testing.

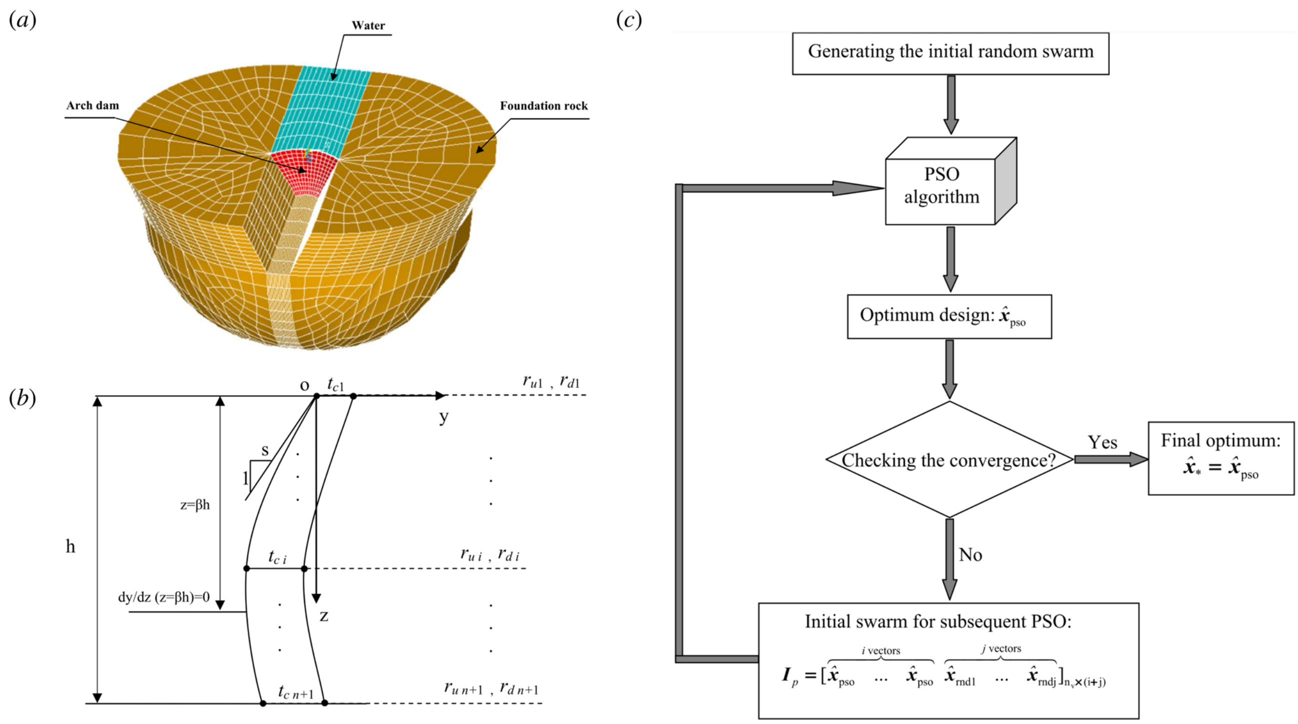

Fig. 15 Dam design and optimisation procedure of Seyedpoor et al. (2012) [90]. **a** Model diagram of Morrow Point arch dam-water-foundation rock system. **b** The central vertical section of arch dam. **c** The framework of the dam design flowchart

Pizarro et al. (2021) [98] presented a structural design platform for reinforced concrete wall buildings, enabling rapid design of wall thickness and length during the initial design process. As presented in Fig. 16, Li et al. (2021) [99] applied AIAD in civil structure optimization to eliminate undesired deformations in energy-absorbing structures. They enhanced crashworthiness and lightweight of structure through accurate deformation mode predictions and multi-objective optimization. Ko et al. (2021) [100] developed a novel approach utilizing ML and knowledge graphs to construct design rules for additive manufacturing, enhancing the understanding of how additive manufacturing processes affect part quality and enabling data-driven prescriptive guidelines.

As scholars' mastery of AI technology improves, the application scenarios of AIAD in civil structures have also been greatly expanded. Latif et al. (2022) [101] employed ML to predict the fundamental period of masonry-infilled reinforced concrete frames. By adopting ML, they realized the integrated optimization of frames, and contributed a predictive dashboard for structural engineering applications. Toosi et al. (2022) [102] developed a Life Cycle Sustainability Assessment (LCSA) model integrated with ML to optimize building design and energy refurbishment, showcasing its effectiveness through a case study on renewable energy-based energy storage systems. Jin et al. (2023) [103] introduced a point-cloud DL method for the inverse design of 3D frame structures via buckling-guided assembly. The inverse design process of Jin et al. is given in Fig. 17, where they realized efficiently maps 3D configurations to 2D precursor patterns, outperforming traditional methods in computational efficiency and applicability to complex shapes. Hu et al. (2023) [104] introduced an ML-based design method for hybrid self-centering braced frames, whose peak interstory drifts and floor accelerations under seismic conditions are controlled by ANN, verifying the effectiveness of the AI approach. Zhao et al. (2024) [105] introduced a GNN-based method for efficient beam layout design in shear wall systems, capable of generating beam layout scenarios by incorporating architectural layouts, design inputs, and engineering experience.

Structural material design is also a pivotal part of civil structures and infrastructures, emphasizing the design and optimization of construction materials to attain better structural properties, durability, and sustainability. The AIAD methodology can realize a thorough and rapid evaluation of material performance under different conditions, thereby accelerating the design cycle. For instance, Sun et al. (2021) [106] developed a Random Forest (RF) and Beetle Antennae search (BAS) algorithm-based ML model for the design and prediction of the mechanical and electrical properties of cementitious composites with conductive fillers, contributing to advancements in intelligent construction practices. Quaranta et al. (2022) [107] integrated mechanics-based formulation with ML techniques

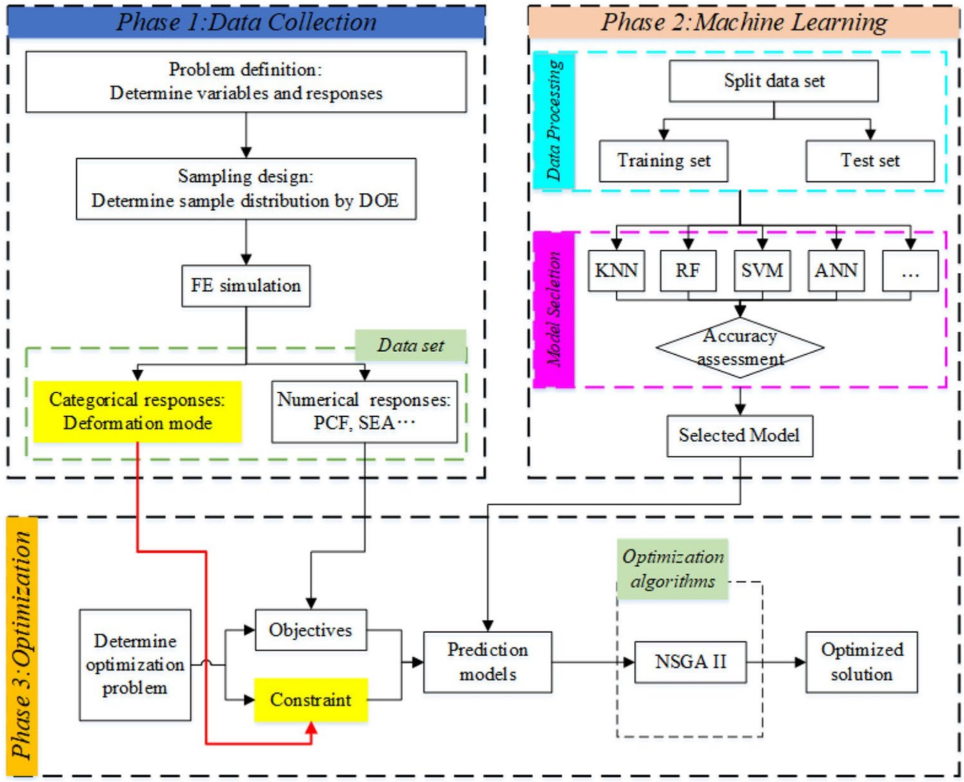

Fig. 16 Schematic diagram of structural design process based on ML of Li et al. [99]

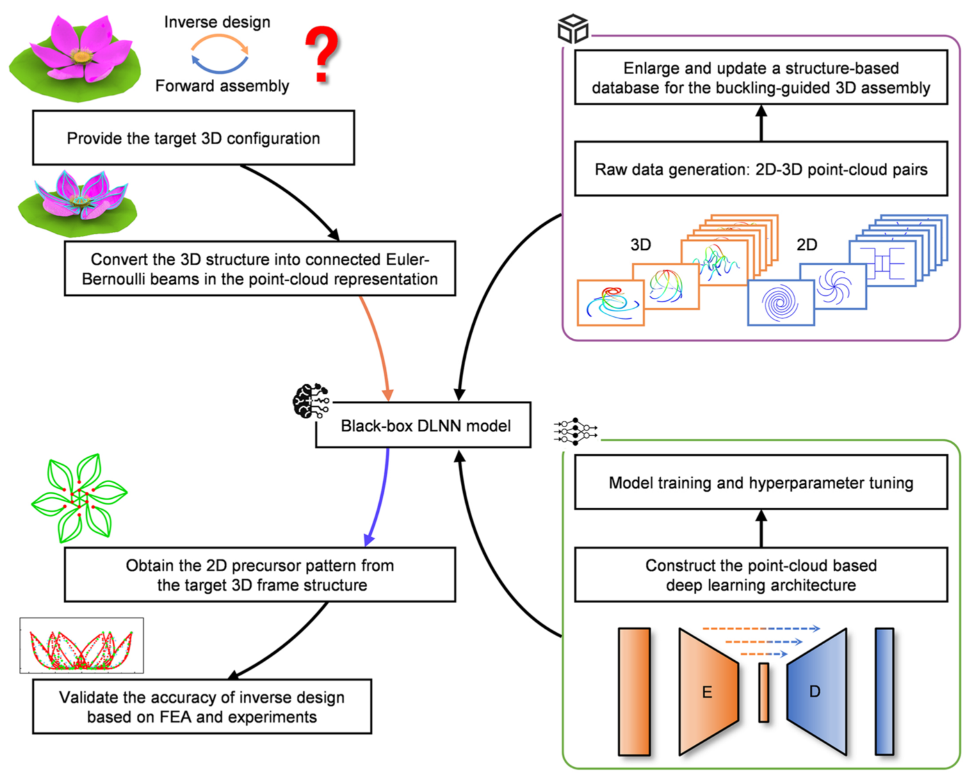

Fig. 17 Schematic diagram of the inverse design workflow of frame structures of Jin et al. (2023) [103]

to enhance shear capacity equations for reinforced concrete elements with stirrups, with the verification case of the application of enhanced equation in structural engineering design. Shafighfard et al. (2022) [108] presented a data-driven approach using stacked ML to predict the compressive strength of steel fiber reinforced concrete under high temperatures, offering a valuable tool for construction material assessment. Geng et al. (2022) [109] developed a data-driven ML model using RF to accurately predict the hardness curve of boron steel, and successfully guided the design of a new press-hardening steel. Figure 18 illustrated a schematic diagram of their work, from data collection to ML model to the final design application Pan et al. (2023) [110] elucidated the potential of ML and AI in the design of high-performance steel materials. They reviewed algorithm selection, model evaluation, and successful cases, focusing on optimizing composition, structure, processing, and performance.

In general, the application of AIAD in the construction of buildings, bridges, and infrastructure projects is revolutionizing the fields of engineering and architecture. By analyzing large datasets to realize fast predicting of the performances of various design alternatives, AIAD is advancing buildings, bridges, and infrastructure by optimizing architectural layouts, improving structural integrity, and enhancing construction materials, while considering the impact of dynamic loads, environmental conditions, and potential failure modes.

7 AI-Aided Design for Topological Structures

In topological structure design, the main target is to find the most efficient distribution of material within a given design space, subject to specified loads and constraints. Traditional topological design approaches depend heavily on manual calculations and iterative processes, which are not only time-consuming but also susceptible to human error. The incorporation of AI technology facilitates the investigation of complex design spaces that conventional methods struggle to navigate, allowing for dynamic adaptation to evolving design requirements and constraints. Such adaptability is essential for creating structures capable of meeting progressively stringent functional demands. Leveraging AI technology, researchers have conducted extensive studies in the domain of topological structure design. Based on the search keywords, the total number of retrieved articles related to the topological structures amounted to 834.

Given the complexity of topological structure design and the limited analytical capabilities of early AI algorithms, initial research efforts were sparse and mostly concentrated on two-dimensional and large-size topological structures. Wu et al. (1994) [111] developed a NN approach to model and classify 2D workpieces based on topological structure, facilitating design and

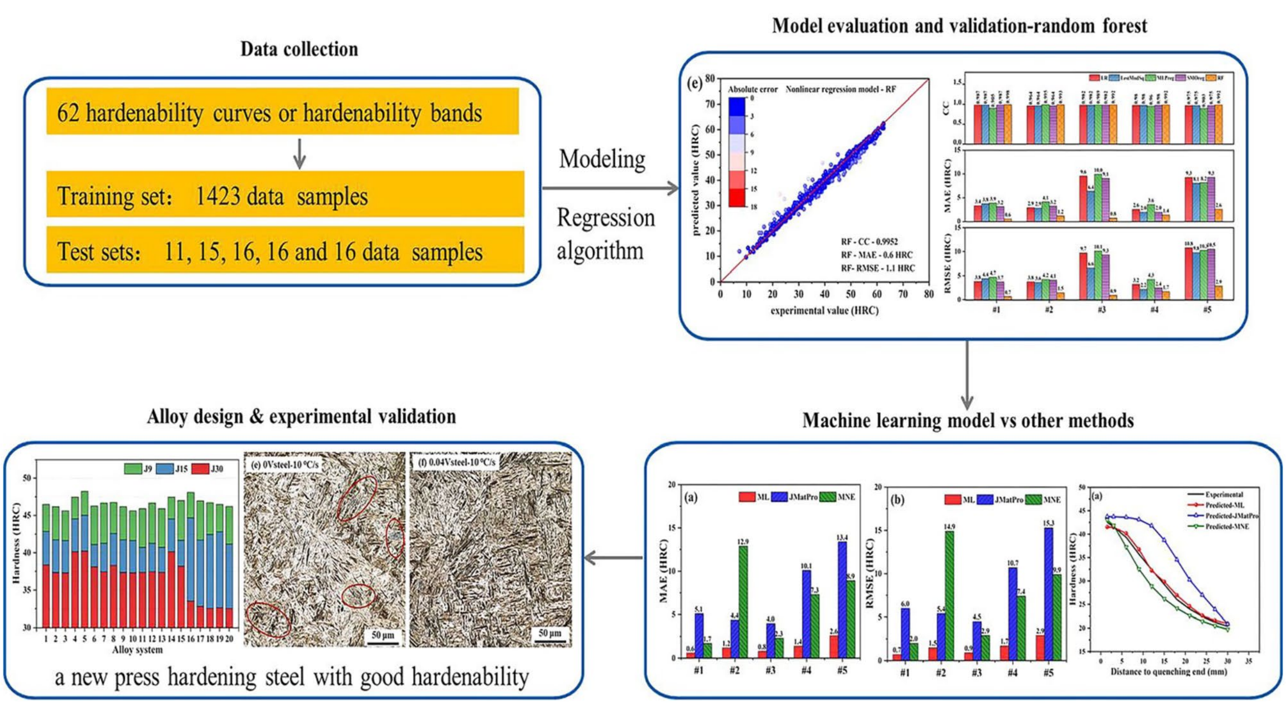

Fig. 18 Schematic diagram of the alloy design workflow of Geng et al. (2023) [109]

manufacturing activities. Hajela and Lee (1997) [112] combined BPNN with a crash response analysis code to aid the design of the rotorcraft subfloor topological structures, as presented in Fig. 19. By adopting GA as the optimization algorithm, they realized the improvement of the energy absorption capacity of the subfloor structure.

Advancements in AI technology have spurred a growing number of researchers to explore AI-aided topological structure design, leading to increasingly complex and realistic application scenarios. Long et al. (2019) [113] adopted ML to design photonic structures with target topological states, enhancing the inverse design of complex material properties. Lee et al. (2022) [114] developed a generative ML algorithm using high-order Bézier curves and a hybrid NN-GA method to design lattice structures with superior mechanical properties, achieving optimized weight-to-performance ratios and validating the results through additive manufacturing and compression testing. The process of lattice structure optimization achieved by Lee et al. based on ML is shown in Fig. 20.

Then, Gao et al. (2023) [115] developed an ML-aided design method for the topology structure of the conformal cooling channels in injection molding, achieving much lower temperature variance and a smaller coolant pressure compared with the conventional conformal cooling design. Cai et al. (2023) [116] introduced DL into the topological photonics system design, obtaining an optimal topological waveguide with reduced propagation loss and achieving a 60% improvement in output. Zheng et al. (2024) [117] introduced an iterative ML approach to accelerate the topological design of compression-only shell structures, enabling real-time assessment of structural performance and construction constraints. The design background and design flowchart in the work of Zheng et al. are depicted in Fig. 21, highlighting the similarities and differences between traditional design methods and ML-based design methods.

AI algorithm advancements have also enhanced the application of AIAD in topological structures with fine and sensitive features, leading to its widespread adoption in microstructure, nanostructures, and metamaterial topological design. Representatively, Xu et al. (2015) [118] developed a ML-based method to identify key microstructure descriptors for designing heterogeneous microstructures, reducing the design space to a manageable size while maintaining essential information. Paul et al. (2019) [119] employed an ML-based feedback-aware data-generation approach to optimize microstructures in Titanium components, providing a vast array of near-optimal solutions that overcome the limitations of conventional methods. Wang et al. (2020) [120] introduced a deep generative modeling method for metamaterial design, realizing properties optimization and graded metamaterial systems creation. Kim et al. (2020) [121] adopted ANN to predict tensile properties of high-strength steels based on microstructural parameters, achieving high accuracy in predicting yield strength, tensile strength, and yield ratio, whose flowchart is given in Fig. 22.

The application of AIAD significantly advanced the design and optimization of microstructure, nanostructures, and metamaterial topology structure, specifically addressing the high costs associated with conventional structure evaluation. Tan et al. (2020) [122] proposed a DL model combining GAN and CNN for the efficient design of microstructural materials, effectively handling geometrical constraints without high-dimensional simulations. Peano et al. (2021) [123] employed DL to classify topological characteristics in periodic nanostructures, significantly accelerating the engineering of domain walls and optimization processes. Peano and Massone (2021) [98] demonstrated how DL can rapidly explore and optimize band structures as well as classify topological properties for arbitrary unit-cell geometries, accelerating design and optimization processes in nanostructure engineering. Ballard et al. (2021) [124] explored the intelligent design of metastructure features of computational sensing systems that optimize data acquisition and enhance sensing capabilities through iterative, data-driven analysis. Naseri et al. (2021) [125] introduced a generative ML-based

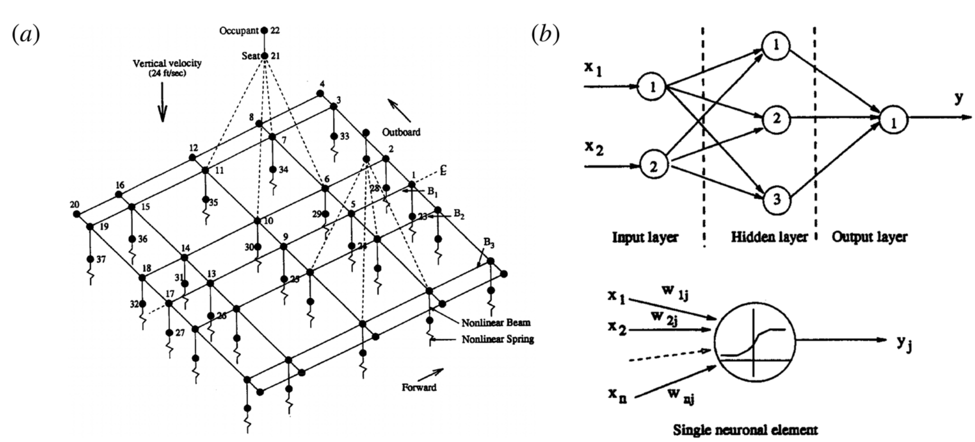

Fig. 19 Subfloor topological structure design and optimization procedure of Hajela and Lee (1997) [112]. **a** KRASH model of the floor section. **b** Architecture of the BPNN

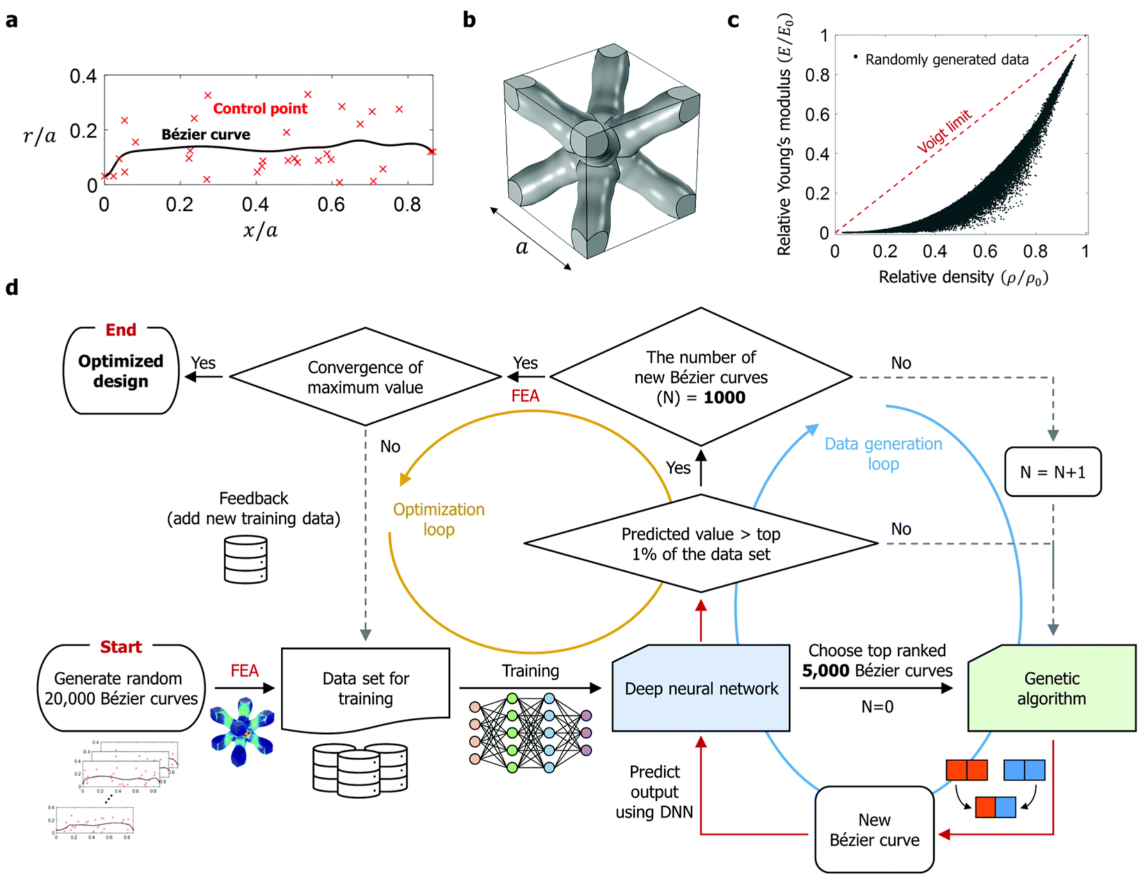

Fig. 20 3D-printed lattice structure optimization aided by active transfer learning of Lee et al. (2022) [114]. **a** The 30th-order Bézier curve with control points where a is the lattice constant. **b** The unit cell of BC structure obtained using Bézier curve. **c** The relative modulus - relative density plot of initial datasets. The initial datasets are obtained from randomly generated control points sets and the relative density has uniform distribution. **d** Workflow chart of the NN-GO design approach

approach for the inverse design of multilayer metasurfaces, automating the synthesis process and optimizing designs to achieve specific scattering properties, avoiding the need for extensive full-wave simulations.

In addition, Seo and Min (2022) [126] introduced a DL-aided approach for multi-scale topology optimization, which efficiently mapped low-dimensional variables to microstructure images, reducing design variables and accelerating optimization processes. Tian et al. (2022) [127] presented an ML-based method for predicting and inversely designing 2D metamaterials with tunable deformation-dependent Poisson's ratio. As shown in Fig. 23, Tian et al. adopted a generative adversarial network (GAN), where the encoder was used to predict the performance of metamaterials, and the decoder was used to perform the performance-based reverse design of metamaterials. Zhang et al. (2023) [128] developed an ML-driven approach for predicting deformations and generating variant designs of auxetic metamaterials, improving upon the traditional FEA-based auxetic metamaterial design method. Liang et al. (2024) [129] introduced a Fourier neural operator-based ML framework for topology design, capable of generating topological structures with clear and smooth boundaries in a short period.

AIAD has been widely used in the design of topological structures. Early, the design target focused on the topology of macrostructures, where the application scenarios were similar to the framework design of civil structures. In recent years, with benefits to the development of DL, especially the GNN, the application scenarios of AIAD in topological structures have mainly transformed into microstructure, nanostructures, and metamaterial. AIAD has greatly advanced the development of these fields by designing or inverse-designing topological of microstructure, nanostructures, and metamaterial.

8 AI-Aided Designs for Composite Structures

Composite structures refer to structures composed of two or more distinct materials. The design process for composite structures necessitates the determination of the matrix

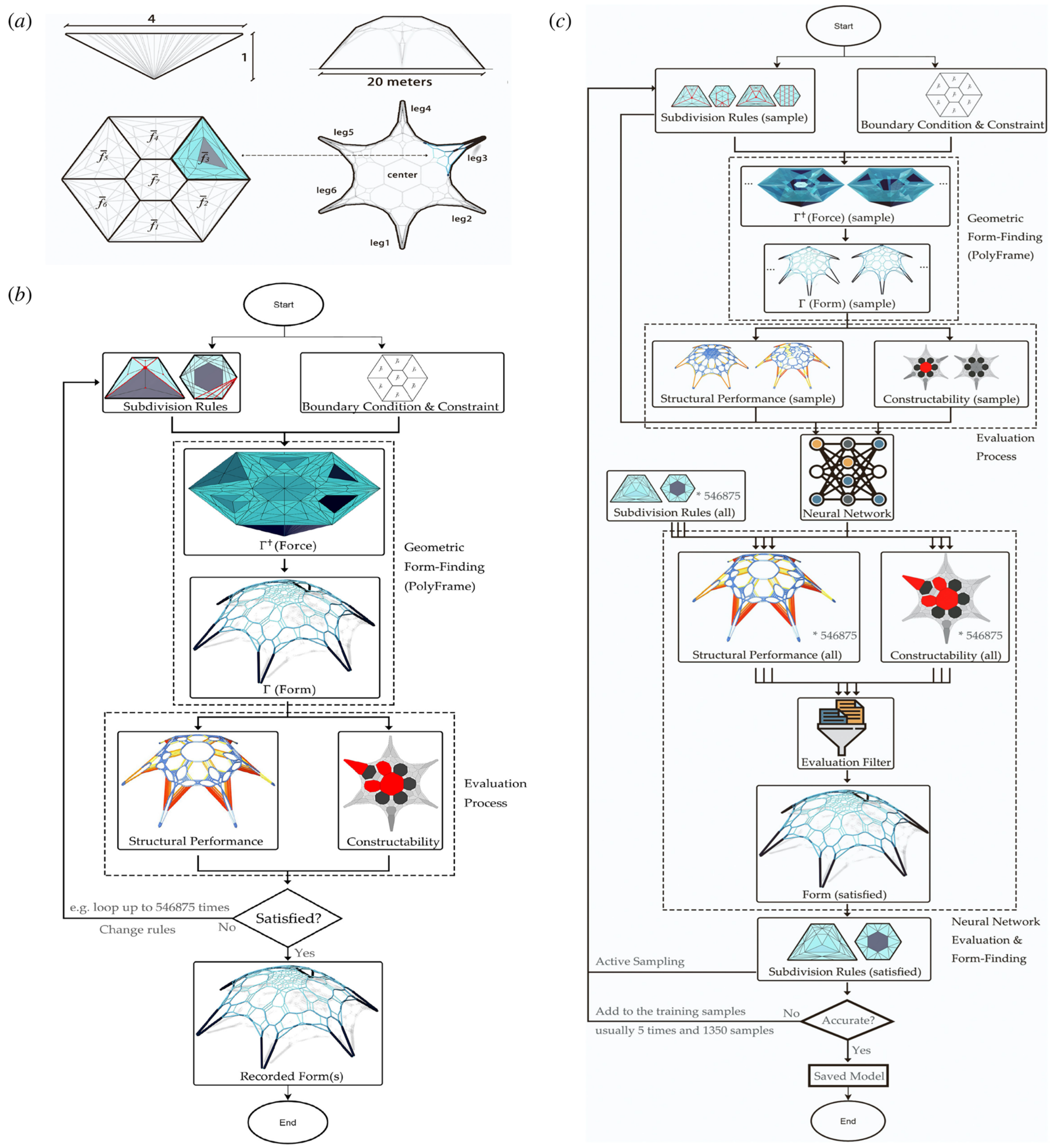

Fig. 21 The shell structures design work of Zheng et al. (2024) [117]. **a** Design background - a single-layer funicular shell structure with six legs across 20 m. **b** Conventional computational form design flowchart for funicular shell structure. **c** Machine learning-assisted funicular shell structure design flowchart

material, reinforcement material, fiber type and orientation, as well as volume fraction. The high dimensionality of design variables and the multiple choices of materials increase the complexity of the entire design process, significantly limiting the development of new composite structures. By adopting AIAD, the lengthy design cycles issue commonly bound with traditional design processes can be addressed, thereby exploring innovative design concepts that might not be readily apparent, and accelerating the overall development of composite structures by streamlining the design-to-production workflow. Researchers have conducted extensive studies of AIAD for composite structures. Based on the search keywords, the total number of retrieved articles amounted to 1541.

The exploration of AI-assisted design methodologies for composite structures primarily commenced in 2004. Ren et al. (2004) [130] used a neural network model to design Young's modulus and cord direction of the tire body rubber-cord composite material layer, realizing the stress of toe optimization. Lew et al. (2004) [131]

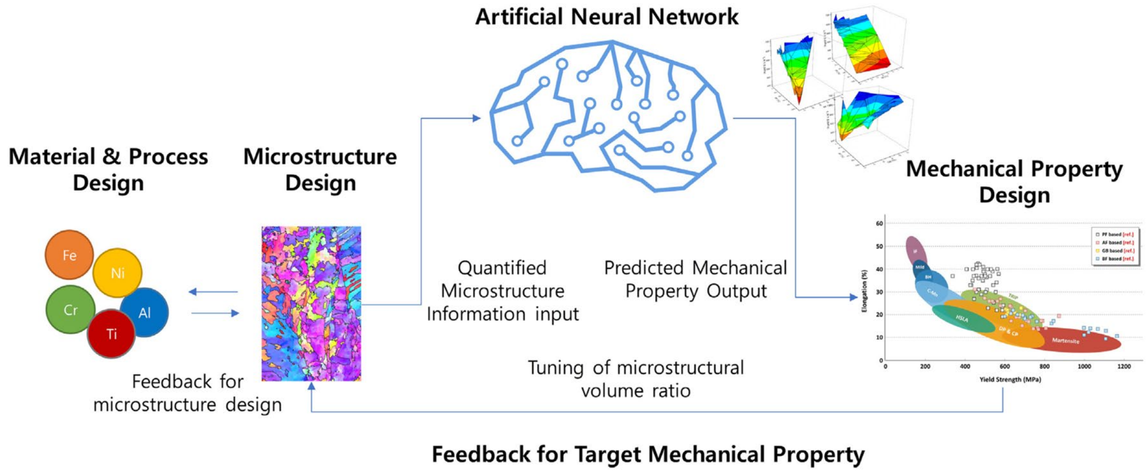

Fig. 22 The flowchart for the ANN-based microstructure design of Kim et al. (2020) [121]

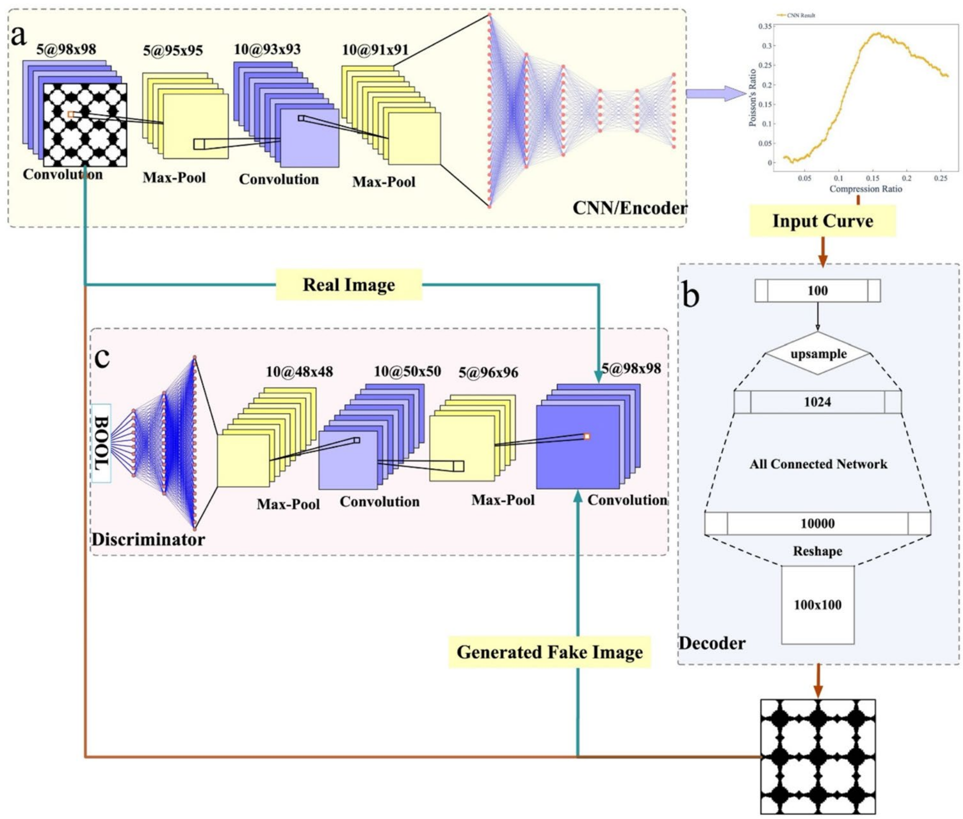

Fig. 23 GAN-based inverse design process of 2D metamaterials of Tian et al. (2022) [127]. **a** CNN-based forward prediction algorithm consisting of a convolutional block and a fully connected block, with input as an image and output as Poisson's ratio-deformation curve. **b** and **c** Cycle-GAN model for inversely designing a 2D metamaterial structure with a designated target Poisson's ratio

proposed an ANN surrogate model to reduce the computational effort involved in homogenization-based composite material modeling. Subsequently, researchers have undertaken a range of studies and investigations. Doreswamy (2008) [132] reviewed Data Mining (DM) frameworks for engineering materials design, proposing their ability to bridge the gap between extracted data and decision-making in composite materials design. Shabani and Mazahery (2012) [133] developed a model combining GA and ANN to optimize matrix composites, confirming its effectiveness in finding optimal process conditions for aluminum matrix composite materials. Gerrard et al. (2014) [134] realized the prediction of bulk Young's modulus and electrical conductivity of two-phase composites based on NNs, highlighting its potential to guide the development of improved structure–property relations.

The emergence of DL has profoundly transformed AI-aided composite structure design, markedly improving predictive precision and optimization processes. The algorithmic advancement enables more effective material selection, structural analysis, and performance forecasting, greatly expanding the exploration ability of researchers, and thereby substantially driving the innovation speed of composite structures. Gu et al. (2018) [135] utilized ML to predict and optimize the mechanical properties of composite materials, significantly enhancing materials' toughness and strength while reducing computational costs. Hamel et al. (2019) [136] developed a ML approach to design active composite structures for 4D printing, optimizing material distribution to achieve precise shape-shifting responses. Gao et al. (2020) [137] employed ML to enhance the design and optimization of conformal porous structures for injection molding, with the result showing a 76% reduction in temperature variance after optimization. Roh et al. (2021) [138] introduced ML algorithms for designing smart, self-sensing fiber-reinforced plastics, optimizing fiber type and electrode distance while considering cost and sensitivity. The flow chart of Roh et al. using ANN to select fiber types can be seen in Fig. 24. Feng (2021) [139] presented an ML-aided stochastic framework for evaluating elastoplastic and damage behaviors in functionally graded structures. By integrating extended support vector regression, they offered a robust method for uncertainty quantification, enhancing structural reliability assessment and design practices in modern composite engineering.

Huang et al. (2021) [140] pioneered an ML-aided design for rubberised concrete, leveraging waste rubber to enhance concrete properties and sustainability. They realized multiple mechanical properties prediction of various concrete mix composites based on ANN, offering a time and cost-efficient approach with reduced carbon footprints. Gao et al. (2023) [115] advanced AI-aided design in manufacturing by applying the ML-aided design method for conformal cooling channels in injection molding. They optimized channel topologies to minimize temperature variance and coolant pressure drop, enhancing production efficiency and part quality. He et al. (2023) [104] reviewed the integration of ML in meta-structure design, highlighting its role in accelerating performance exploration and optimization across acoustic, elastic, and mechanical domains, underscoring ML's potential to revolutionize traditional design methods. Okafor et al. (2023) [141] summarized the recent advances in ML-aided design of reinforced polymer composite and hybrid material systems. They also summarized the data-driven composite design route in their work, outlining the production routes, data acquisition sources, and simulation tools used during the ML integration process, which is shown in Fig. 25. Lee et al. (2023) [142] reviewed various ML algorithms for materials design, offering guidelines for selecting models based on material characteristics, with a focus on the inverse design of complex composite structures.

Similar to topological structure design, the application of AIAD in composite structure design facilitates the exploration of new structures in this field. The application of AIAD greatly reduces design costs and offers robust support for scholars to explore new materials, processes, and structural forms.

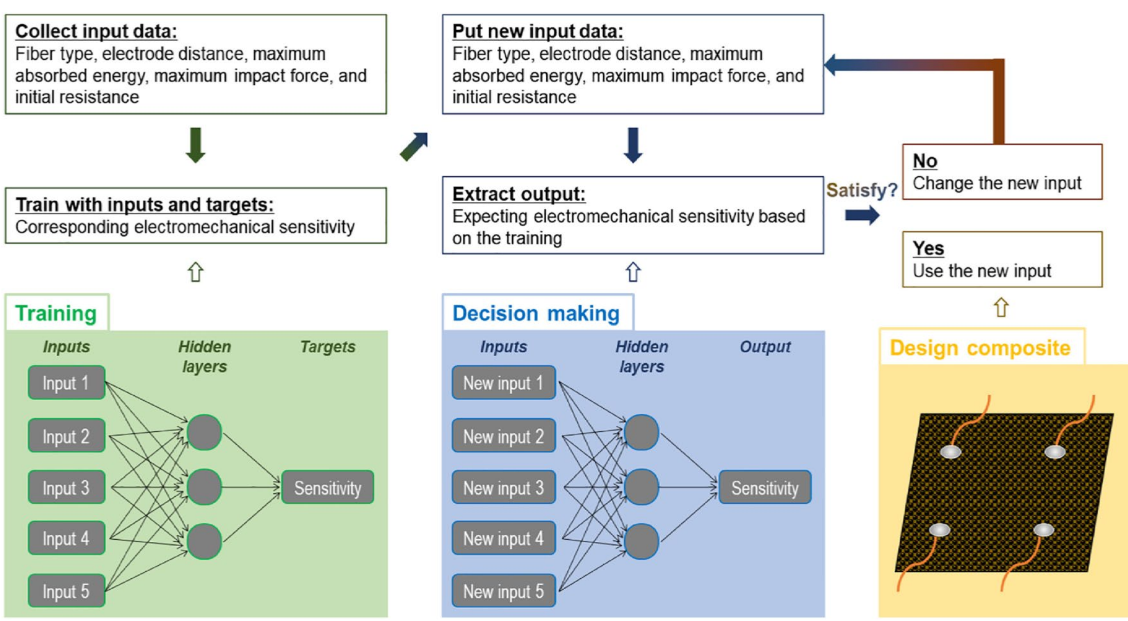

Fig. 24 ANN-based electromechanical sensitivity optimization process of Roh et al. (2021) [138]

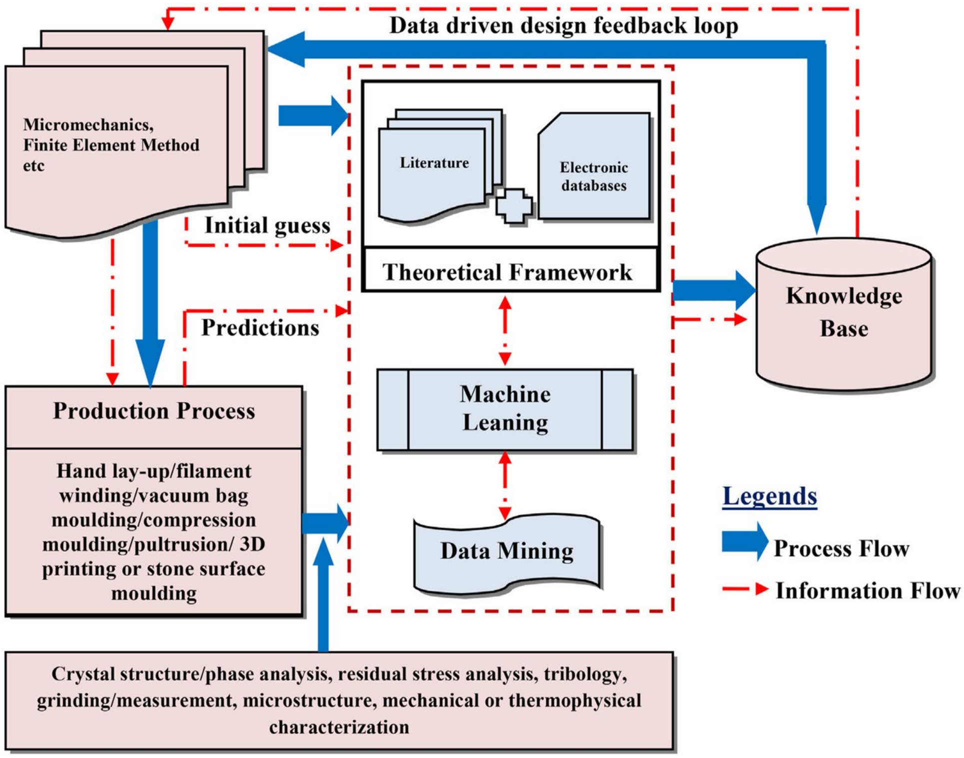

Fig. 25 Data-driven design of composite material process of Okafor et al. (2023) [141]

9 Discussions, Challenges, and Future Perspectives

9.1 Discussions

By reviewing the diverse applications of AI-aided design (AIAD) in the context of complex engineering systems, it is evident that AI technology has been extensively applied to structural design across a broad spectrum of fields. These fields encompass macrostructural design, such as shipbuilding, aerospace, automotive, and civil structures, as well as microstructural design, including topological and composite structures. Figure 26 provided a detailed summary of the application scenarios of AIAD for structures across domains.

As shown in Fig. 26, the application of AIAD in macrostructures primarily focuses on optimizing design parameters, enhancing overall performance, and reducing construction costs. For instance, in shipbuilding structures, AIAD is

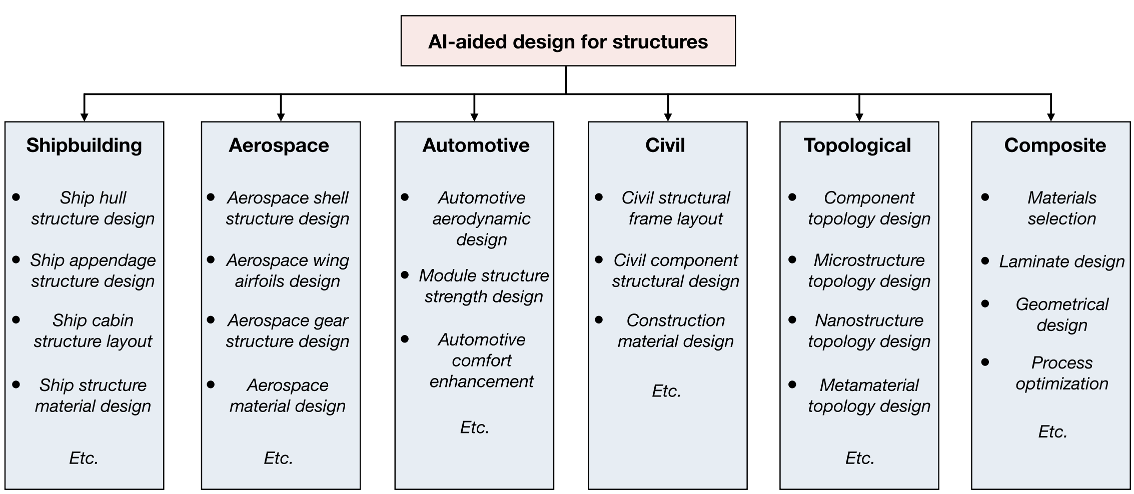

Fig. 26 Application scenarios of AIAD for structures across domains

used to optimize hydrodynamic performance by deforming ship hull structures. In aerospace structures, AIAD is widely used in the structural layout and wing shape design. In automotive structures, AIAD is used to reduce structural weight and improve the driving experience. In civil structures, AIAD optimizes the design of bridges, buildings, and other infrastructures by predicting potential structural failures and optimizing material use.

For microstructural applications, AIAD is widely used for structure exploration and inverse design. In topological structures, AIAD facilitates the exploration of topological solutions by revealing hidden patterns in the design space, leading to the development of microstructures or metamaterials with improved mechanical properties. In composite structures, AIAD optimizes material composition and arrangement to achieve desired mechanical properties and enhance the reliability and performance of composite materials under various conditions.

9.2 Main Challenges

Although AIAD has proven its capabilities in structural design, it still mainly remains in academic research and is subject to certain limitations when applied to actual engineering projects. The main reasons may be threefold: data requirements, algorithm credibility, and industry acceptance.

1. Data requirements: When applying AIAD, a substantial amount of high-quality structural design data are required, which should not only encompass a wide range of design cases but also include detailed design variables and performance metrics. However, in most situations, obtaining such comprehensive data can be challenging in certain specific domains. In some cases, the workload required to acquire data can even exceed that of traditional structural design methods, thereby decreasing the practicality of AIAD.
2. Algorithm credibility: Although AIAD performs well in experimental and simulated environments, its reliability and robustness in real-world applications still require further validation. In complex and dynamic real-world settings, AI algorithms may encounter unforeseen conditions and challenges, necessitating assurance of accuracy evaluation of the design scheme under diverse conditions. This demands extensive testing and validation efforts to guarantee the credibility of the algorithms, which limits the application of AIAD.
3. Industry acceptance: Existing structural design methods and standards have been widely accepted and utilized. Integrating AIAD technology into these frameworks requires overcoming user acceptance in both technical and usability aspects. Particularly, the commonly used ML and DL algorithms in pattern regression exhibit black-box characteristics, leading to potential skepticism among users and designers regarding the design outcomes. Consequently, when designing costly and complex structures, industries still tend to rely on and trust traditional methods, limiting the further application of AIAD in structural design.

9.3 Future Perspectives

In the past, due to algorithmic limitations and a lack of data, AIAD could only address idealized and simplified structural design problems, making it difficult to apply in actual engineering projects. Today, benefiting from the continuous optimization of AI algorithms and the rapid enhancement of computer performance, AIAD has gradually been accepted by engineers and designers, becoming a widely used tool in structural design.

In the future, with ongoing advancements in algorithms and computing capabilities, AIAD will find further applications in structural design. The current challenges faced by AIAD, such as data requirements, algorithm reliability, and industry acceptance, will also be progressively overcome as the credibility and robustness of AI algorithms continue to improve. In addition, the advantages of AIAD's knowledge decoupling capabilities will be further reflected in the future, making it easier for designers in different fields to cooperate with each other and realize multidisciplinary structural design. Ultimately, AIAD will have the capability to be safely applied to more real-world engineering structural design projects.

10 Conclusions

In this article, we have provided a comprehensive and state-of-the-art review of applications of AIAD to structures across various disciplines, including naval architecture and marine engineering, specifically shipbuilding design, aerospace, automotive, civil, mechanical and topological optimization, and composite structures. By adopting AI algorithms, AIAD can achieve design knowledge decoupling and design labor reduction, solving the knowledge-intensive and labor-intensive challenges. Extensive application examples demonstrated that AIAD can not only shorten the design cycle for structural design problems from macrostructure to microstructure but also effectively solve the high-dimensional, multi-objective, and multi-constraint challenges encountered in the modern structural design process. Although AIAD currently faces a series of challenges such as data requirements, algorithm reliability, and industry acceptance, we believe that with the continuous advancement of AI technology and computing power, the application of AIAD in structural design will be greatly expanded, driving the intelligent, efficient, and creative design of future structures.

Author Contributions A.Y. and S.L. wrote the main manuscript text and H.D. edited the manuscript. A.Y. and S.L. prepare the figures. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Gellatly R, Berke L (1971) Optimal structural design, vol 70. Air Force Flight Dynamics Laboratory, Air Force Systems Command, Wright-Patterson Air Force Base
2. Banichuk NV (2013) Problems and methods of optimal structural design, vol 26. Springer, New York
3. Bayazit N (2004) Investigating design: a review of forty years of design research. Des Issues 20(1):16–29
4. Guo X, Cheng GD (2010) Recent development in structural design and optimization. Acta Mech Sin 26(6):807–823
5. Deb K, Sindhya K, Hakanen J (2016) Multi-objective optimization. In: Decision sciences. CRC Press, Boca Raton, pp 161–200
6. Winston PH (1984) Artificial intelligence. Addison-Wesley Longman, Boston
7. Nilsson NJ (1982) Principles of artificial intelligence. Springer, Berlin
8. Turing AM (2009) Computing machinery and intelligence. Springer, Berlin
9. Moor J (2006) The Dartmouth College artificial intelligence conference: the next fifty years. AI Mag 27(4):87–87
10. Waterman DA (1985) A guide to expert systems. Addison-Wesley Longman, Reading
11. Lloyd JW (2012) Foundations of logic programming. Springer, Berlin
12. Murphy KP (2012) Machine learning: a probabilistic perspective. MIT, Cambridge
13. Mahesh B (2020) Machine learning algorithms-a review. Int J Sci Res (IJSR) 9(1):381–386
14. Jordan MI, Mitchell TM (2015) Machine learning: trends, perspectives, and prospects. Science 349(6245):255–260
15. Caruana R, Niculescu-Mizil A (2006) An empirical comparison of supervised learning algorithms. In: Proceedings of the 23rd international conference on machine learning, pp 161–168
16. Hastie T, Tibshirani R, Friedman J, Hastie T, Tibshirani R, Friedman J (2009) Overview of supervised learning. In: The elements of statistical learning: data mining, inference, and prediction. Springer, pp 9–41
17. Barlow HB (1989) Unsupervised learning. Neural Comput 1(3):295–311
18. Zhu XJ (2005) Semi-supervised learning literature survey. Technical Report 1530, Computer Science, University of Wisconsin-Madison
19. Kaelbling LP, Littman ML, Moore AW (1996) Reinforcement learning: a survey. J Artif Intell Res 4:237–285
20. Rumelhart DE, Hinton GE, Williams RJ (1986) Learning representations by back-propagating errors. Nature 323(6088):533–536
21. Vanluchene RD, Sun R (1990) Neural networks in structural engineering. Comput Aided Civ Infrastruct Eng 5(3):207–215
22. Hajela P, Berke L (1991) Neurobiological computational models in structural analysis and design. Comput Struct 41(4):657–667
23. Ghaboussi J, Garrett JH Jr, Wu X (1991) Knowledge-based modeling of material behavior with neural networks. J Eng Mech 117(1):132–153
24. Reich Y, Konda SL, Levy SN, Monarch IA, Subrahmanian E (1993) New roles for machine learning in design. Artif Intell Eng 8(3):165–181
25. Kang HT, Yoon CJ (1994) Neural network approaches to aid simple truss design problems. Comput Aided Civ Infrastruct Eng 9(3):211–218
26. Adeli H, Park HS (1995) A neural dynamics model for structural optimization-theory. Comput Struct 57(3):383–390
27. Mukherjee A, Deshpande JM (1995) Modeling initial design process using artificial neural networks. J Comput Civ Eng 9(3):194–200
28. Goel V, Chen J (1996) Application of expert network for material selection in engineering design. Comput Ind 30(2):87–101
29. Reich Y (1997) Machine learning techniques for civil engineering problems. Comput Aided Civ Infrastruct Eng 12(4):295–310
30. MacCallum KJ (1990) Does intelligent cad exist? Artif Intell Eng 5(2):55–64
31. Duffy AHB, Persidis A, MacCallum KJ (1996) Nodes: a numerical and object based modelling system for conceptual engineering design. Knowl Based Syst 9(3):183–206
32. Duffy SM, Duffy AHB, MacCallum KJ (1995) A design reuse model. In: Proceedings of the international conference on engineering design (ICED '95), pp 490–495
33. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521(7553):436–444
34. Goodfellow I, Bengio Y, Courville A (2016) Deep learning. MIT, Cambridge
35. Seuring S, Gold S (2012) Conducting content-analysis based literature reviews in supply chain management. Supply Chain Manag 17(5):544–555
36. Ray T, Gokarn RP, Sha OP (1996) Neural network applications in naval architecture and marine engineering. Artif Intell Eng 10(3):213–226
37. Yang JB, Sen P (1997) Multiple attribute design evaluation of complex engineering products using the evidential reasoning approach. J Eng Des 8(3):211–230
38. Reich Y (1998) Learning in design: from characterizing dimensions to working systems. AI EDAM 12(2):161–172
39. Alkan AD, Gülez K (2004) A knowledge-based computational design tool for determining preliminary stability particulars of naval ships. Nav Eng J 116(4):37–52
40. Ramazani MR, Noroozi S, Koohgilani M, Cripps B, Sewell P (2013) Determination of the static pressure loads on a marine composite panel from strain measurements utilising artificial neural networks. Proc Inst Mech Eng Part M 227(1):12–21

41. Li B, Pang YJ, Cheng YX, Zhu XM (2017) Collaborative optimization for ring-stiffened composite pressure hull of underwater vehicle based on lamination parameters. Int J Nav Architect Ocean Eng 9(4):373–381
42. Ahn Y, Kim Y, Kim SY (2019) Database of model-scale sloshing experiment for LNG tank and application of artificial neural network for sloshing load prediction. Mar Struct 66:66–82
43. Ao Y, Li Y, Gong J, Li S (2023) An artificial intelligence-aided design (AIAD) of ship hull structures. J Ocean Eng Sci 8(1):15–32
44. Ao Y, Xu J, Zhang D, Li S (2024) Artificial intelligence aided design (AIAD) of hull form of unmanned underwater vehicles (UUVs) for minimization of energy consumption. J Comput Inf Sci Eng 24(1):1–28
45. Yu A, Li Y, Li S, Gong J (2024) Construction high precision neural network proxy model for ship hull structure design based on hybrid datasets of hydrodynamic loads. J Mar Sci Appl 23(4):1–15
46. Ao Y, Duan H, Li S (2024) An integrated-hull design assisted by artificial intelligence-aided design method. Comput Struct 297:107320
47. Huang L, Pena B, Liu Y, Anderlini E (2022) Machine learning in sustainable ship design and operation: a review. Ocean Eng 266:112907
48. Romero-Tello P (2023) Prediction of seakeeping in the early stage of conventional Monohull vessels design using artificial neural network. J Ocean Eng Sci 8(4):344–366
49. Romero-Tello P, Gutierrez-Romero JE, Servan-Camas B (2023) Generation of machine learning algorithms for the calculation of a ship's response amplitude operators. SSRN. https://doi.org/10.2139/ssrn.4634746
50. Romero-Tello P, Gutierrez-Romero JE, Servan-Camas B (2024) Development of artificial neural networks to predict the response amplitude operators for seakeeping of ships navigating at forward speed. SSRN. https://doi.org/10.2139/ssrn.4707086
51. Bagazinski NJ, Ahmed F (2023) Shipgen: a diffusion model for parametric ship hull generation with multiple objectives and constraints. J Mar Sci Eng 11(12):2215
52. Bagazinski NJ, Ahmed F (2023) Ship-d: ship hull dataset for design optimization using machine learning. In: International design engineering technical conferences and computers and information in engineering conference, vol 87301. American Society of Mechanical Engineers, p V03AT03A028
53. Tripathi S, Rajagopalan V (2023) Deep neural network (DNN) assisted parametric optimisation of stern flap for a high speed displacement ship. In: ISOPE international ocean and polar engineering conference
54. Umaretiya JR, Joshi SP (1992) An insight into the expert-seisd: a knowledge based system for structural design. Eng Comput 8:151–161
55. Rokhsaz K, Steck JE (1993) Application of artificial neural networks in nonlinear aerodynamics and aircraft design. Technical report, SAE Technical Paper
56. Giannakoglou KC (2002) Design of optimal aerodynamic shapes using stochastic optimization methods and computational intelligence. Prog Aerosp Sci 38(1):43–76
57. Forrester AI, Bressloff NW, Keane AJ (2006) Optimization using surrogate models and partially converged computational fluid dynamics simulations. Proc R Soc A Math Phys Eng Sci 462(2071):2177–2204
58. Park KH, Jun SO, Baek SM, Cho MH, Yee KJ, Lee DH (2013) Reduced-order model with an artificial neural network for aerostructural design optimization. J Aircraft 50(4):1106–1116
59. Lindhorst K, Haupt MC, Horst P (2014) Efficient surrogate modelling of nonlinear aerodynamics in aerostructural coupling schemes. AIAA J 52(9):1952–1966
60. Garriga AG, Mainini L, Ponnusamy SS (2019) A machine learning enabled multi-fidelity platform for the integrated design of aircraft systems. J Mech Des 141(12):121405
61. Pitton SF, Ricci S, Bisagni C (2019) Buckling optimization of variable stiffness cylindrical shells through artificial intelligence techniques. Compos Struct 230:111513
62. Dong Y, Tao J, Zhang Y, Lin W, Ai J (2021) Deep learning in aircraft design, dynamics, and control: review and prospects. IEEE Trans Aerosp Electron Syst 57(4):2346–2368
63. Agarwal KM, Tyagi RK, Saxena KK (2022) Deformation analysis of al alloy aa2024 through equal channel angular pressing for aircraft structures. Adv Mater Process Technol 8(1):828–842
64. Yong-fei J, Guo-shuai N, Yang Y, Yong-bing D, Jiao Z, Yan-feng H, Bao-de S (2023) Knowledge-aware design of high-strength aviation aluminum alloys via machine learning. J Market Res 24:346–361
65. Dede G (2024) Machine learning-aided generative design methodology for a Martian regolith habitation shell. Adv Space Res 73(8):3909–3935
66. Al-Haddad LA, Mahdi NM (2024) Efficient multidisciplinary modeling of aircraft undercarriage landing gear using data-driven naïve Bayes and finite element analysis. Multiscale Multidiscip Model Exp Des 7(4):1–13
67. Li J, Zhang M (2021) Data-based approach for wing shape design optimization. Aerosp Sci Technol 112:106639
68. Renganathan SA, Maulik R, Ahuja J (2021) Enhanced data efficiency using deep neural networks and Gaussian processes for aerodynamic design optimization. Aerosp Sci Technol 111:106522
69. Liu B, Liang H, Han ZH, Yang G (2022) Surrogate-based aerodynamic shape optimization of a morphing wing considering a wide Mach-number range. Aerosp Sci Technol 124:107557
70. Wu P, Yuan W, Ji L, Zhou L, Zhou Z, Feng W, Guo Y (2022) Missile aerodynamic shape optimization design using deep neural networks. Aerosp Sci Technol 126:107640
71. Taj ZUD, Bilal A, Awais M, Salamat S, Abbas M, Maqsood A (2023) Design exploration and optimization of aerodynamics and radar cross section for a fighter aircraft. Aerosp Sci Technol 133:108114
72. Zhang Z, Ao Y, Li S, Gu GX (2024) An adaptive machine learning-based optimization method in the aerodynamic analysis of a finite wing under various cruise conditions. Theor Appl Mech Lett 14(1):100489
73. Sharma RS, Hosder S (2024) Mission-driven inverse design of blended wing body aircraft with machine learning. Aerospace 11(2):137
74. Karali H, Inalhan G, Tsourdos A (2024) AI-driven multidisciplinary conceptual design of unmanned aerial vehicles. In: AIAA SCITECH 2024 Forum, p 1708
75. Susca L, Mandorli F, Rizzi C, Cugini U (2000) Racing car design using knowledge aided engineering. AI EDAM 14(3):235–249
76. Bae D, Jung W, Sohn I (2003) A fatigue design method of spot-welded lap joint using neural network. Int J Mod Phys B 17(08n09):1684–1690
77. Yao B, Feng T (2018) Machine learning in automotive industry. Adv Mech Eng. https://doi.org/10.1177/1687814018805787
78. Wang S, Cheng Q, Wang L, Xu J, Fang X, Kang J (2022) Research on the rapid design technology of automotive inspection structure based on MBD model. Proc Inst Mech Eng Part D J Automob Eng 236(8):1670–1686
79. Kim S, Jwa M, Lee S, Park S, Kang N (2022) Deep learning-based inverse design for engineering systems: multidisciplinary design optimization of automotive brakes. Struct Multidiscip Optim 65(11):323

80. Wang M, Jia S, Chen E, Yang S, Liu P, Qi Z (2022) Research and application of neural network for tread wear prediction and optimization. Mech Syst Signal Process 162:108070
81. Al-Shabi MA, Obaideen K, Nassif AB, Bettayeb M (2023) Design car side impact using machine learning. In: Unmanned systems technology XXV, vol 12549. SPIE, pp 121–128
82. Töpel E, Fuchs A, Büttner K, Kaliske M, Prokop G (2023) Machine-learning-based design optimization of chassis bushings. Vehicles 6(1):1–21
83. Cheng Q, Zhao Y, Zhuang J, Alshamrani AM (2024) Characterizing the time-dependent external force on the cars' hood door in accident using deep neural networks. Mater Today Commun 38:107587
84. Cao X, Wang Q, Li H, Ma H (2023) Research on fast CFD simulation of automobile flow field based on artificial intelligence. J Phys Conf Ser 2441:012011
85. Elrefaie M, Dai A, Ahmed F (2024) Drivaernet: a parametric car dataset for data-driven aerodynamic design and graph-based drag prediction. arXiv preprint. arXiv:2403.08055
86. Ma L, Wu Y, Li Q, Yuan X (2022) Recognition of car front facing style for machine-learning data annotation: a quantitative approach. Symmetry 14(6):1181
87. Ding F, Xie W, Xie X (2023) Research on optimization of car door closing sound quality based on the integration of structural simulation and test. Proc Inst Mech Eng Part D 237(6):1378–1390
88. Miles JC, Moore CJ (1991) An expert system for the conceptual design of bridges. Comput Struct 40(1):101–105
89. Reich Y, Fenves SJ (1995) System that learns to design cable-stayed bridges. J Struct Eng 121(7):1090–1100
90. Seyedpoor SM, Salajegheh J, Salajegheh E (2012) Shape optimal design of materially nonlinear arch dams including dam-water-foundation rock interaction using an improved pso algorithm. Optim Eng 13(1):79–100
91. Akinosho TD, Oyedele LO, Bilal M, Ajayi AO, Delgado MD, Akinade OO, Ahmed AA (2020) Deep learning in the construction industry: a review of present status and future innovations. J Build Eng 32:101827
92. Xie Y, Ebad Sichani M, Padgett JE, DesRoches R (2020) The promise of implementing machine learning in earthquake engineering: a state-of-the-art review. Earthquake Spectra 36(4):1769–1801
93. Málaga-Chuquitaype C (2022) Machine learning in structural design: an opinionated review. Front Built Environ 8:815717
94. Afshari Sajad Saraygord, Enayatollahi Fatemeh, Xiangyang Xu, Liang Xihui (2022) Machine learning-based methods in structural reliability analysis: a review. Reliab Eng Syst Saf 219:108223
95. Ocak A, Nigdeli SM, Bekdaş G, Işıkdağ Ü (2023) Artificial intelligence and deep learning in civil engineering. In: Hybrid metaheuristics in structural engineering: including machine learning applications. Springer, pp 265–288
96. Geyer P, Singaravel S (2018) Component-based machine learning for performance prediction in building design. Appl Energy 228:1439–1453
97. Liu F, Jiang X, Wang X, Wang L (2020) Machine learning-based design and optimization of curved beams for multistable structures and metamaterials. Extreme Mech Lett 41:101002
98. Pizarro PN, Massone LM (2021) Structural design of reinforced concrete buildings based on deep neural networks. Eng Struct 241:112377
99. Li Z, Ma W, Yao S, Xu P, Hou L, Deng G (2021) A machine learning based optimization method towards removing undesired deformation of energy-absorbing structures. Struct Multidiscip Optim 64:919–934
100. Ko H, Witherell P, Lu Y, Kim Sn, Rosen DW (2021) Machine learning and knowledge graph based design rule construction for additive manufacturing. Addit Manuf 37:101620
101. Latif I, Banerjee A, Surana M (2022) Explainable machine learning aided optimization of masonry infilled reinforced concrete frames. Structures 44:1751–1766
102. Toosi HA, Lavagna M, Leonforte F, Del Pero C, Aste N (2022) A novel LCSA-machine learning based optimization model for sustainable building design-a case study of energy storage systems. Build Environ 209:108656
103. Jin T, Cheng X, Xu S, Lai Y, Zhang Y (2023) Deep learning aided inverse design of the buckling-guided assembly for 3d frame structures. J Mech Phys Solids 179:105398
104. Hu S, Wang W, Alam MS, Zhu S, Ke K (2023) Machine learning-aided peak displacement and floor acceleration-based design of hybrid self-centering braced frames. J Build Eng 72:106429
105. Zhao P, Liao W, Huang Y, Lu X (2024) Beam layout design of shear wall structures based on graph neural networks. Autom Constr 158:105223
106. Sun J, Ma Y, Li J, Zhang J, Ren Z, Wang X (2021) Machine learning-aided design and prediction of cementitious composites containing graphite and slag powder. J Build Eng 43:102544
107. Quaranta G, De Domenico D, Monti G (2022) Machine-learning-aided improvement of mechanics-based code-conforming shear capacity equation for RC elements with stirrups. Eng Struct 267:114665
108. Shafighfard T, Bagherzadeh F, Rizi RA, Yoo DY (2022) Data-driven compressive strength prediction of steel fiber reinforced concrete (SFRC) subjected to elevated temperatures using stacked machine learning algorithms. J Market Res 21:3777–3794
109. Geng X, Cheng Z, Wang S, Peng C, Ullah A, Wang H, Wu G (2022) A data-driven machine learning approach to predict the hardenability curve of boron steels and assist alloy design. J Mater Sci 57(23):10755–10768
110. Pan G, Wang F, Shang C, Wu H, Wu G, Gao J, Wang S, Gao Z, Zhou X, Mao X (2023) Advances in machine learning-and artificial intelligence-assisted material design of steels. Int J Miner Metall Mater 30(6):1003–1024
111. Wu MC, Chen JR, Jen SR (1994) Global shape information modelling and classification of 2d workpieces. Int J Comput Integr Manuf 7(5):261–275
112. Hajela P, Lee E (1997) Topological optimization of rotorcraft subfloor structures for crashworthiness considerations. Comput Struct 64(1–4):65–76
113. Long Y, Ren J, Li Y, Chen H (2019) Inverse design of photonic topological state via machine learning. Appl Phys Lett 114(18):181105
114. Lee S, Zhang Z, Gu GX (2022) Generative machine learning algorithm for lattice structures with superior mechanical properties. Mater Horiz 9(3):952–960
115. Gao Z, Dong G, Tang Y, Zhao YF (2023) Machine learning aided design of conformal cooling channels for injection molding. J Intell Manuf 34(5):1–19
116. Cai C, Li X, He G, Lian F, Li M, Wang Q, Qin Y (2023) Inverse design of dual-band photonic topological insulator beam splitters for efficient light transmission. J Phys D Appl Phys 57(13):135301
117. Zheng H, Moosavi V, Akbarzadeh M (2020) Machine learning assisted evaluations in structural design and construction. Autom Constr 119:103346
118. Xu H, Liu R, Choudhary A, Chen W (2015) A machine learning-based design representation method for designing heterogeneous microstructures. J Mech Des 137(5):051403

119. Paul A, Acar P, Liao W, Choudhary A, Sundararaghavan V, Agrawal A (2019) Microstructure optimization with constrained design objectives using machine learning-based feedback-aware data-generation. Comput Mater Sci 160:334–351
120. Wang L, Chan YC, Ahmed F, Liu Z, Zhu P, Chen W (2020) Deep generative modeling for mechanistic-based learning and design of metamaterial systems. Comput Methods Appl Mech Eng 372:113377
121. Kim D, Lee J, Lee MS, Son HJ, Reddy NS, Kim M, Moon SK, Kim KT, Yu JH, Kim S (2020) Artificial intelligence for the prediction of tensile properties by using microstructural parameters in high strength steels. Materialia 11:100699
122. Tan RK, Zhang NL, Ye W (2020) A deep learning-based method for the design of microstructural materials. Struct Multidiscip Optim 61:1417–1438
123. Peano V, Sapper F, Marquardt F (2021) Rapid exploration of topological band structures using deep learning. Phys Rev X 11(2):021052
124. Ballard Z, Brown C, Madni AM, Ozcan A (2021) Machine learning and computation-enabled intelligent sensor design. Nat Mach Intell 3(7):556–565
125. Naseri P, Hum SV (2021) A generative machine learning-based approach for inverse design of multilayer metasurfaces. IEEE Trans Antennas Propag 69(9):5725–5739
126. Seo M, Min S (2022) Novel material representation method via a deep learning model for multi-scale topology optimization. Adv Eng Softw 174:103300
127. Tian J, Tang K, Chen X, Wang X (2022) Machine learning-based prediction and inverse design of 2d metamaterial structures with tunable deformation-dependent Poisson's ratio. Nanoscale 14(35):12677–12691
128. Zhang C, Ridard A, Kibsey M, Zhao YF (2023) Variant design generation and machine learning aided deformation prediction for auxetic metamaterials. Mech Mater 181:104642
129. Liang K, Zhu D, Li F (2024) A Fourier neural operator-based lightweight machine learning framework for topology optimization. Appl Math Model 129:714–732
130. Ren XC, Yao Z (2004) Structure optimization of pneumatic tire using an artificial neural network. In: International symposium on neural networks. Springer, pp 841–847
131. Lew TL, Scarpa F, Worden K (2004) Homogenisation metamodelling of perforated plates. Strain 40(3):103–112
132. Doreswamy (2008) A survey for data mining frame work for polymer matrix composite engineering materials design applications. Int J Comput Intell Syst 1(4):313–328
133. Shabani MO, Mazahery A (2012) Optimization of process conditions in casting aluminum matrix composites via interconnection of artificial neurons and progressive solutions. Ceram Int 38(6):4541–4547
134. Gerrard DD, Fullwood DT, Halverson DM (2014) Correlating structure topological metrics with bulk composite properties via neural network analysis. Comput Mater Sci 91:20–27
135. Gu GX, Chen CT, Buehler MJ (2018) De novo composite design based on machine learning algorithm. Extreme Mech Lett 18:19–28
136. Hamel Craig M, Roach Devin J, Long Kevin N, Demoly Frédéric, Dunn Martin L, Jerry Qi H (2019) Machine-learning based design of active composite structures for 4d printing. Smart Mater Struct 28(6):065005
137. Gao Z, Sossou D, Zhao YF (2020) Machine learning aided design and optimization of conformal porous structures. In: International design engineering technical conferences and computers and information in engineering conference, vol 83983. American Society of Mechanical Engineers, p V009T09A041
138. Roh HD, Lee D, Lee IY, Park YB (2021) Machine learning aided design of smart, self-sensing fiber-reinforced plastics. Composites Part C 6:100186
139. Feng Y (2021) Machine learning aided stochastic elastoplastic and damage analysis of functionally graded structures. PhD thesis, UNSW Sydney
140. Huang X, Zhang J, Sresakoolchai J, Kaewunruen S (2021) Machine learning aided design and prediction of environmentally friendly rubberised concrete. Sustainability 13(4):1691
141. Okafor CE, Iweriolor S, Ani OI, Ahmad S, Mehfuz S, Ekwueme GO, Chukwumuanya OE, Abonyi SE, Ekengwu IE, Chikelu OP (2023) Advances in machine learning-aided design of reinforced polymer composite and hybrid material systems. Hybrid Adv 2:100026
142. Lee J, Park D, Lee M, Lee H, Park K, Lee I, Ryu S (2023) Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: a review. Mater Horiz. https://doi.org/10.1039/D3MH00039G